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(54) **OPTICAL WAVEGUIDE SENSOR**

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(57) **ABSTRACT**

An optical waveguide sensor includes: a plurality of first waveguides, each first waveguide including a core at least partially enclosed by a cladding and being configured to guide an examination light; and a contact surface with which an examination subject is in contact and from which leaks seeping component of the examination light guided by the first waveguides. The examination light guided by the plurality of first waveguides has mutually different amount of leakage from the contact surface.

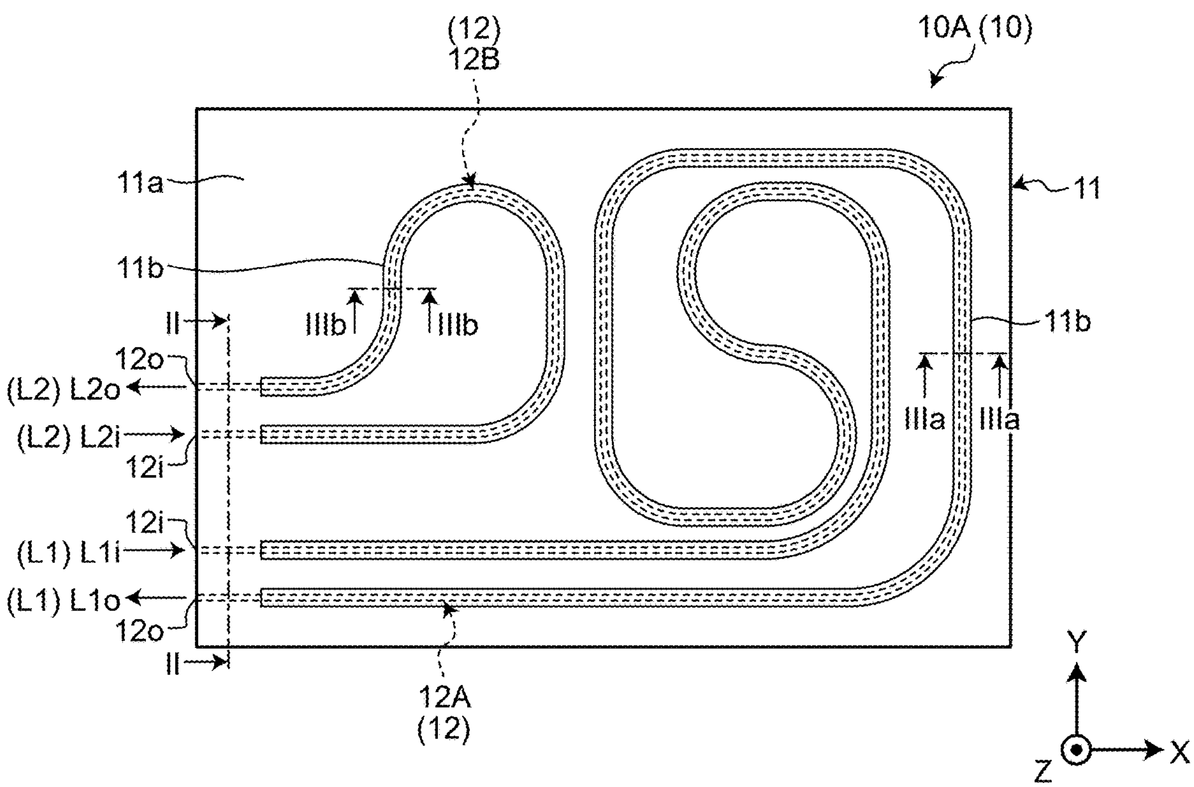


FIG.1

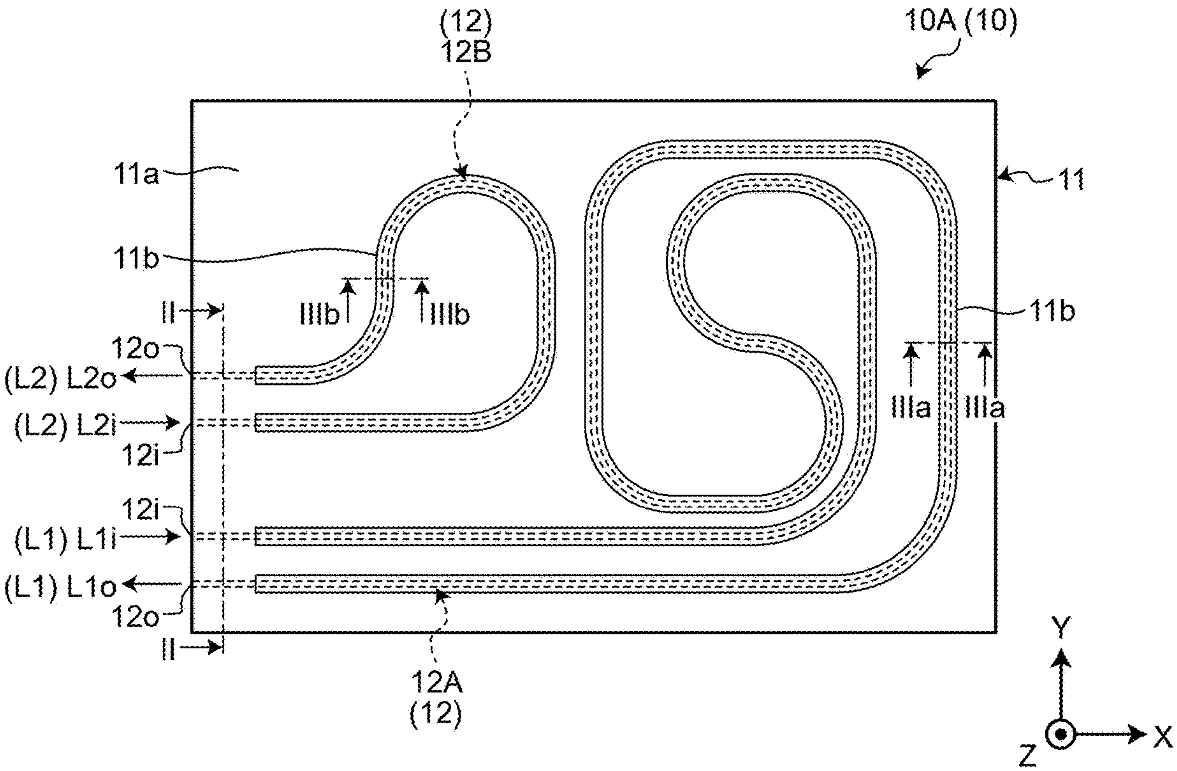

(12)
12B
10A (10)
11a
11
11b
IIIb
IIIb
11b
II
12o
(L2) L2o
(L2) L2i
12i
IIIa
IIIa
12i
(L1) L1i
(L1) L1o
12o
II
12A
(12)
Y
Z
X

FIG.2

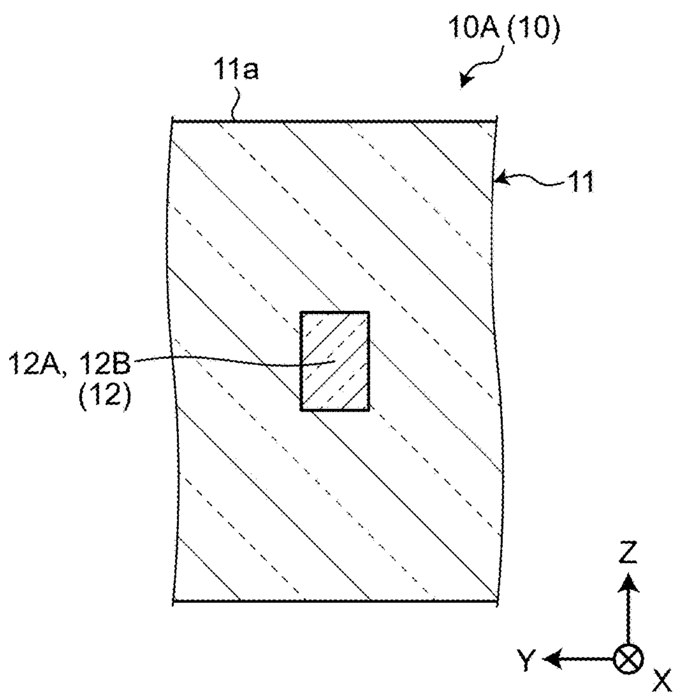

10A (10)
11a
11
12A, 12B
(12)
Z
Y
X

FIG.3

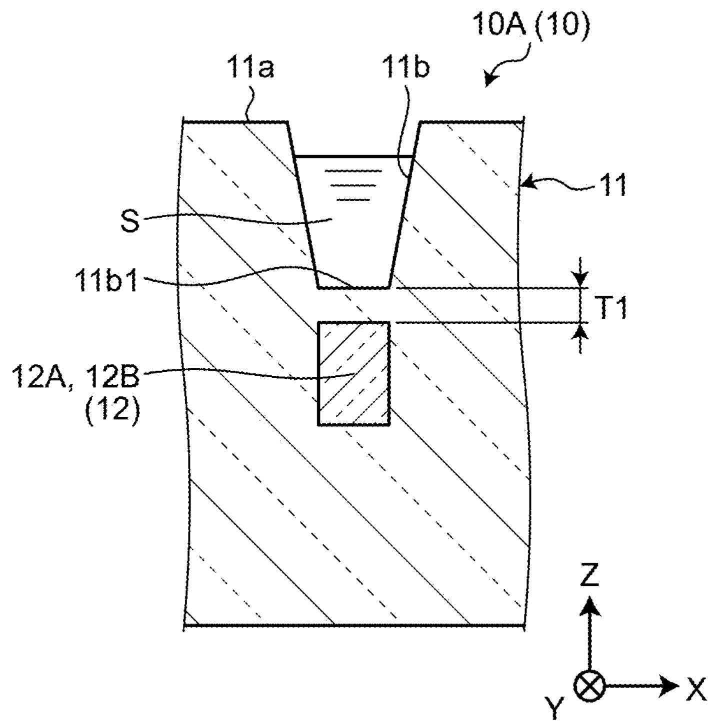


FIG.4

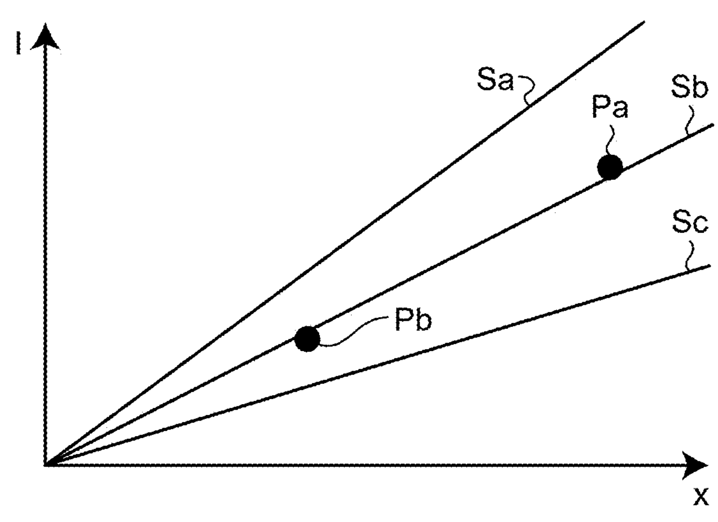
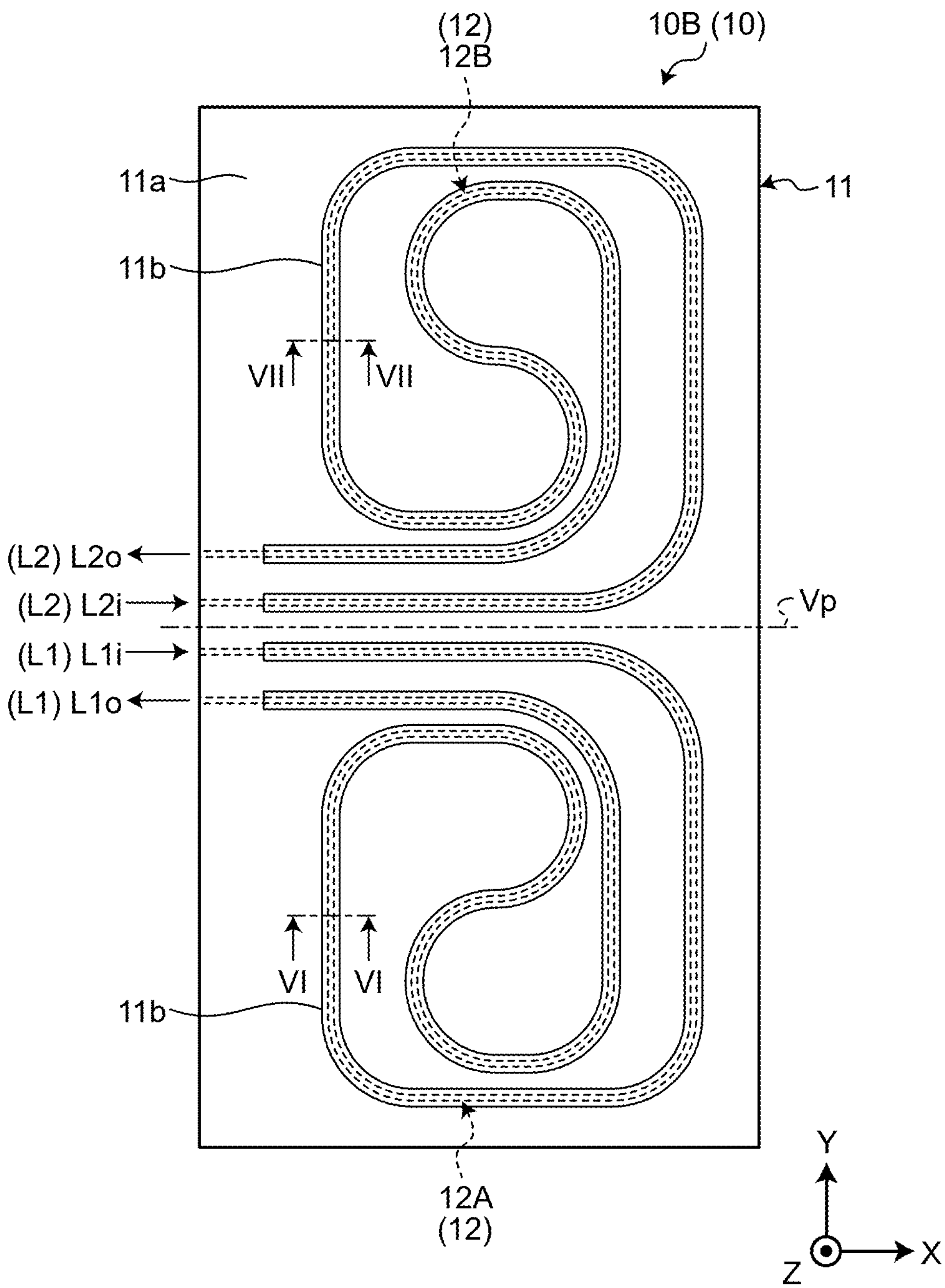


FIG.5



(12)
12B
10B (10)
11a
11
11b
VII
VII
(L2) L2o
(L2) L2i
Vp
(L1) L1i
(L1) L1o
VI
VI
11b
12A
(12)
Y
X
Z

FIG.6

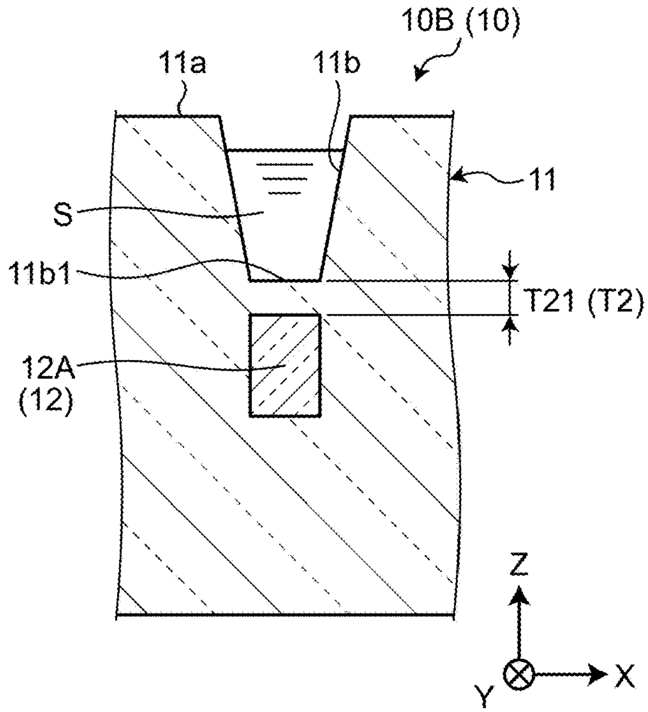

10B (10)
11a
11b
11
S
11b1
T21 (T2)
12A
(12)
Z
Y
X

FIG.7

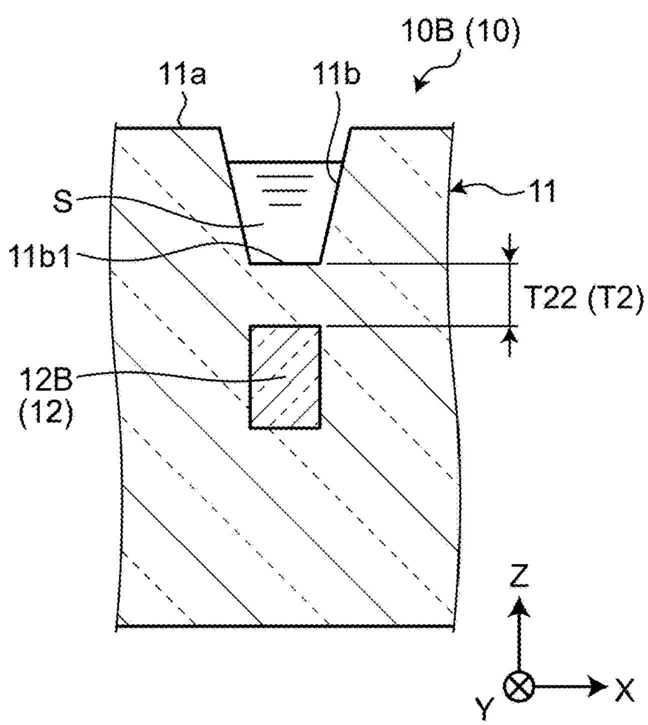
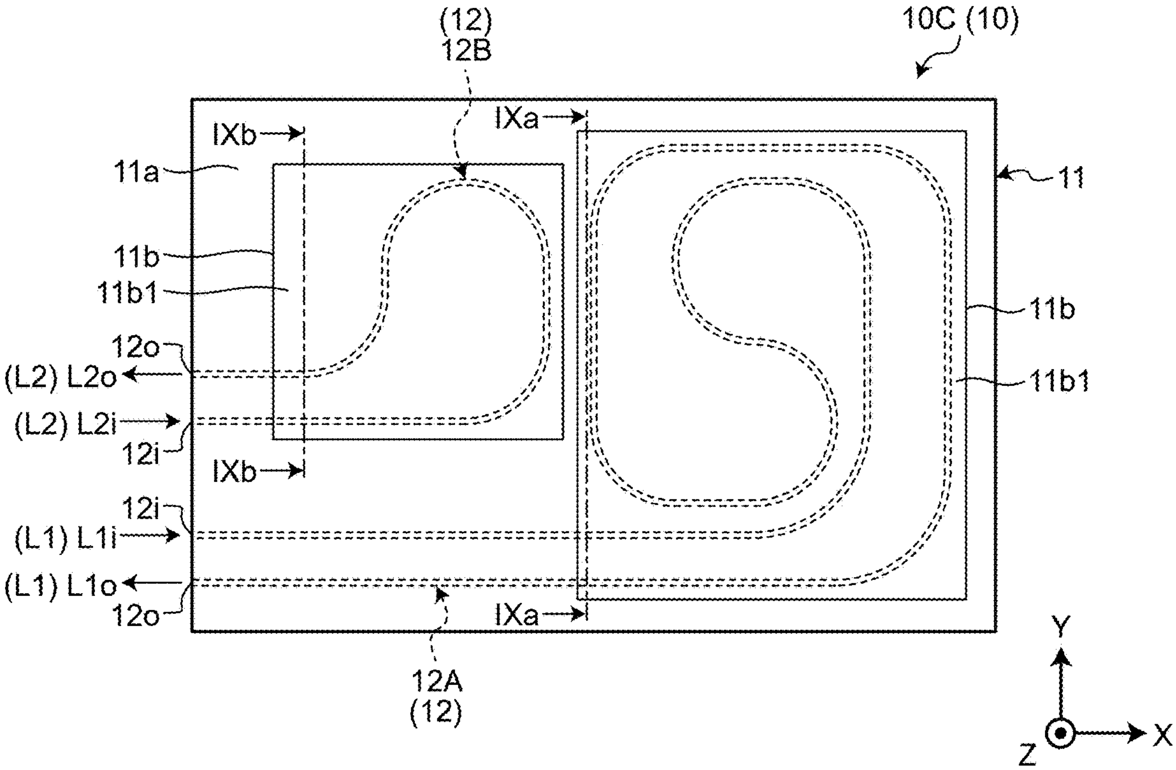
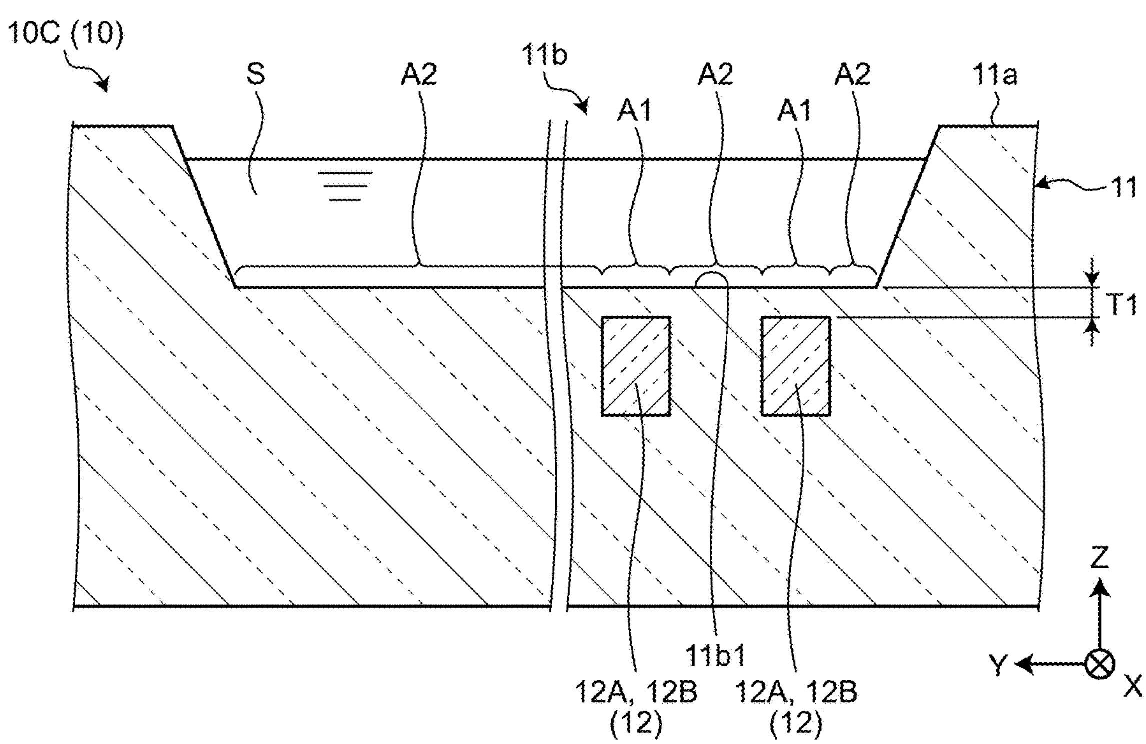

10B (10)
11a
11b
11
S
11b1
T22 (T2)
12B
(12)
Z
Y
X

FIG.8



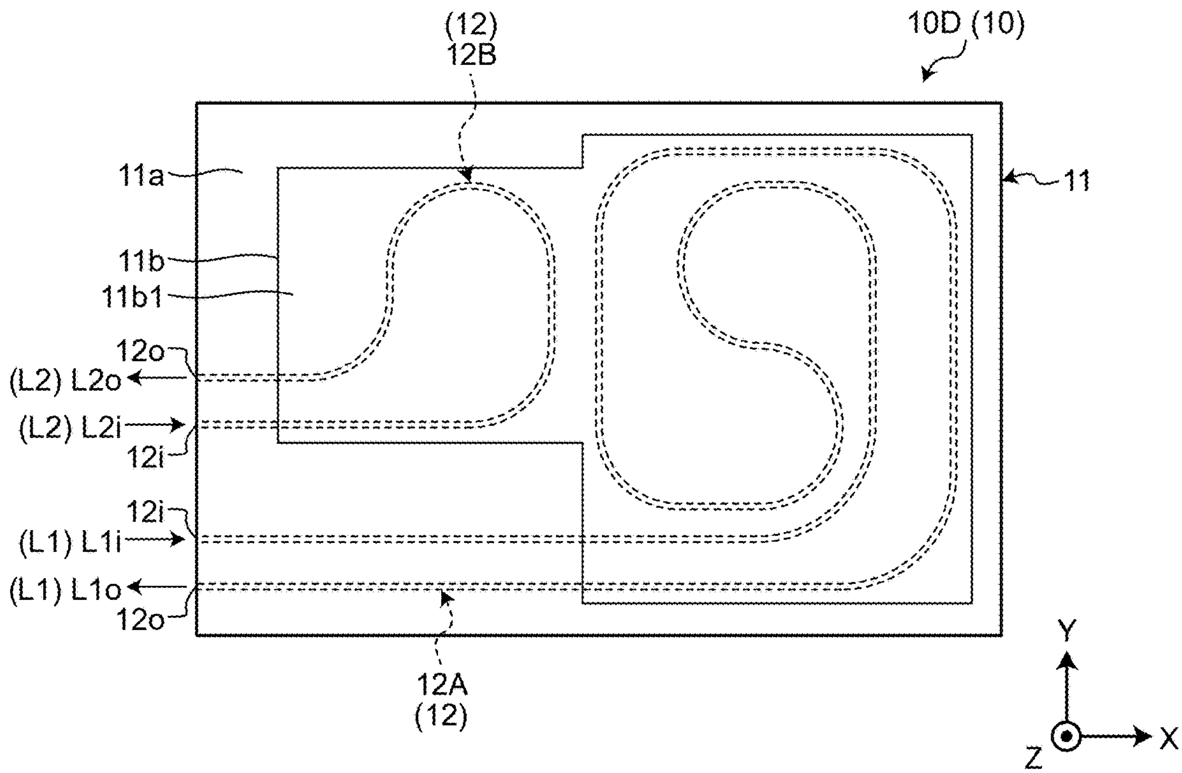
(12)
12B
10C (10)
IXa
IXb
11a
11
11b
11b1
12o
(L2) L2o
(L2) L2i
12i
IXb
11b
11b1
12i
(L1) L1i
(L1) L1o
12o
IXa
12A
(12)
Y
Z
X

FIG.9



10C (10)
S
A2
11b
A1
A2
A1
A2
11a
11
T1
11b1
12A, 12B
(12)
12A, 12B
(12)
Z
Y
X

FIG.10



(12)
12B
10D (10)
11a
11
11b
11b1
12o
(L2) L2o
(L2) L2i
12i
12i
(L1) L1i
(L1) L1o
12o
12A
(12)
Y
X
Z

FIG.11

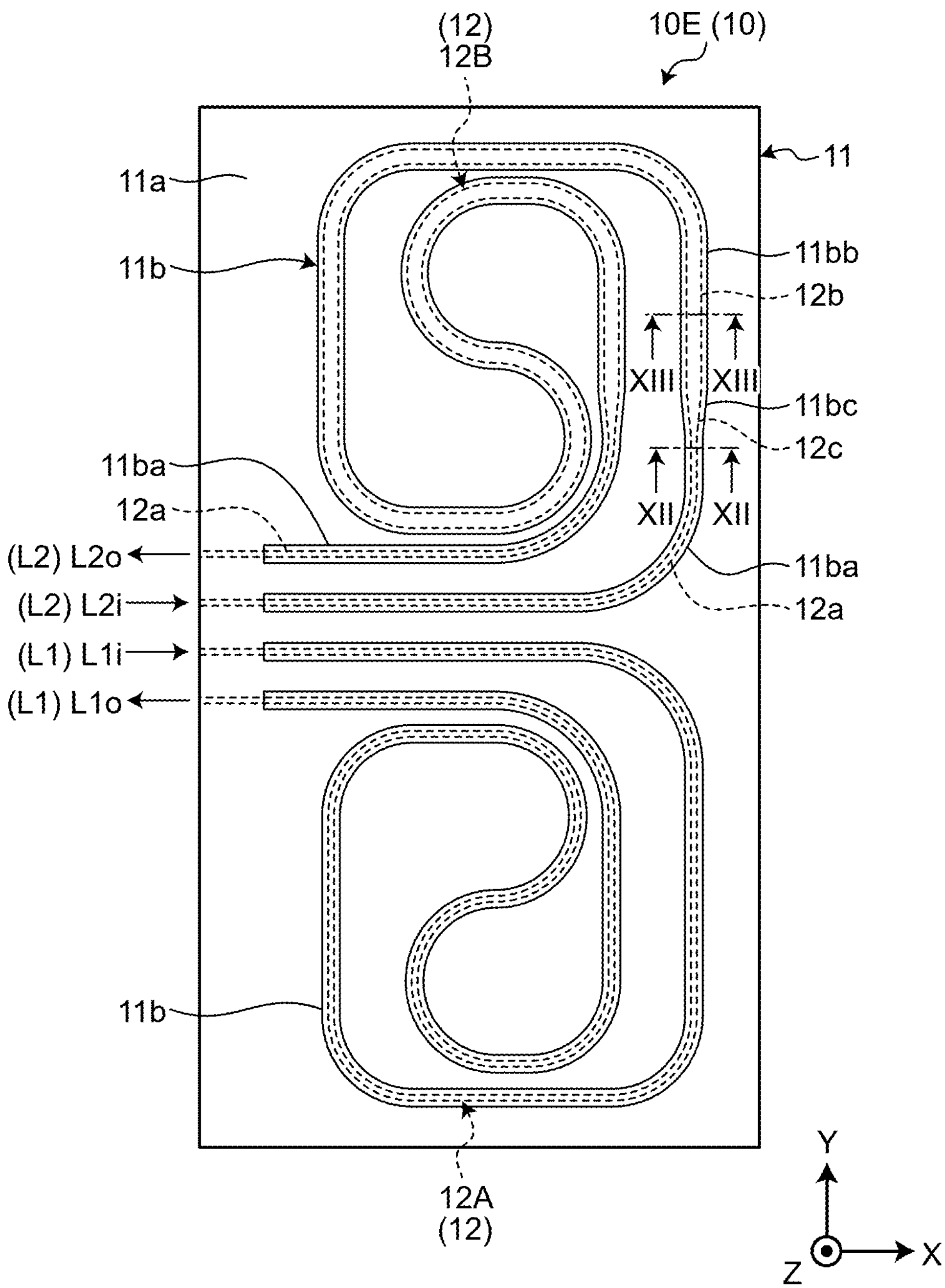
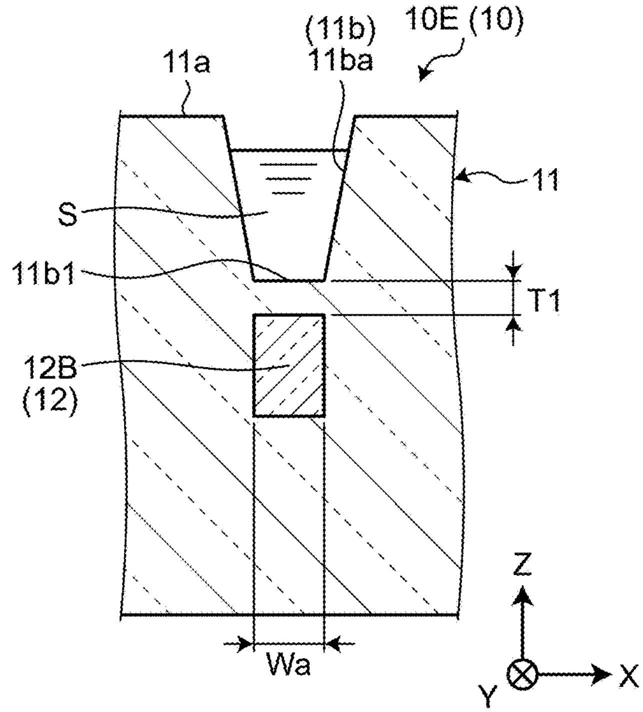
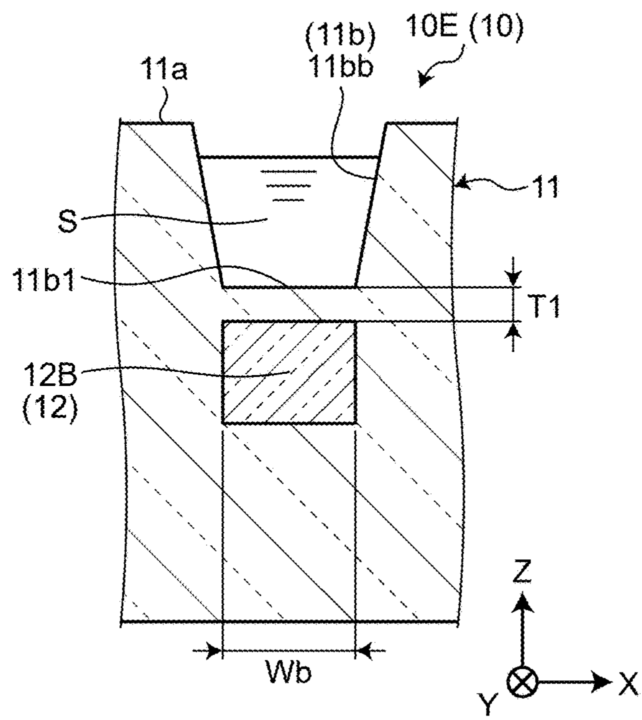

(12)
12B
10E (10)
11
11a
11b
11bb
12b
XIII
XIII
11bc
12c
11ba
12a
XII
XII
(L2) L2o
(L2) L2i
(L1) L1i
(L1) L1o
11ba
12a
11b
12A
(12)
Y
X
Z

FIG.12



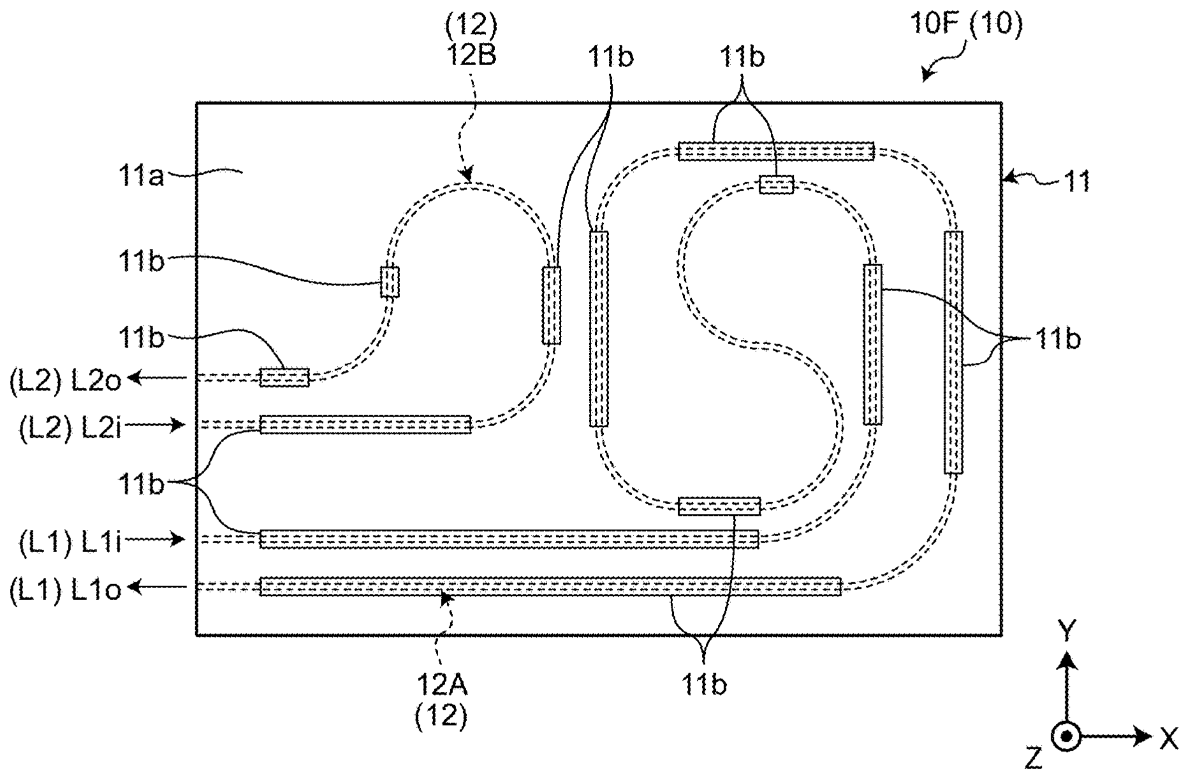
10E (10)
(11b)
11ba
11a
11
S
11b1
T1
12B
(12)
Wa
Z
Y
X

FIG.13



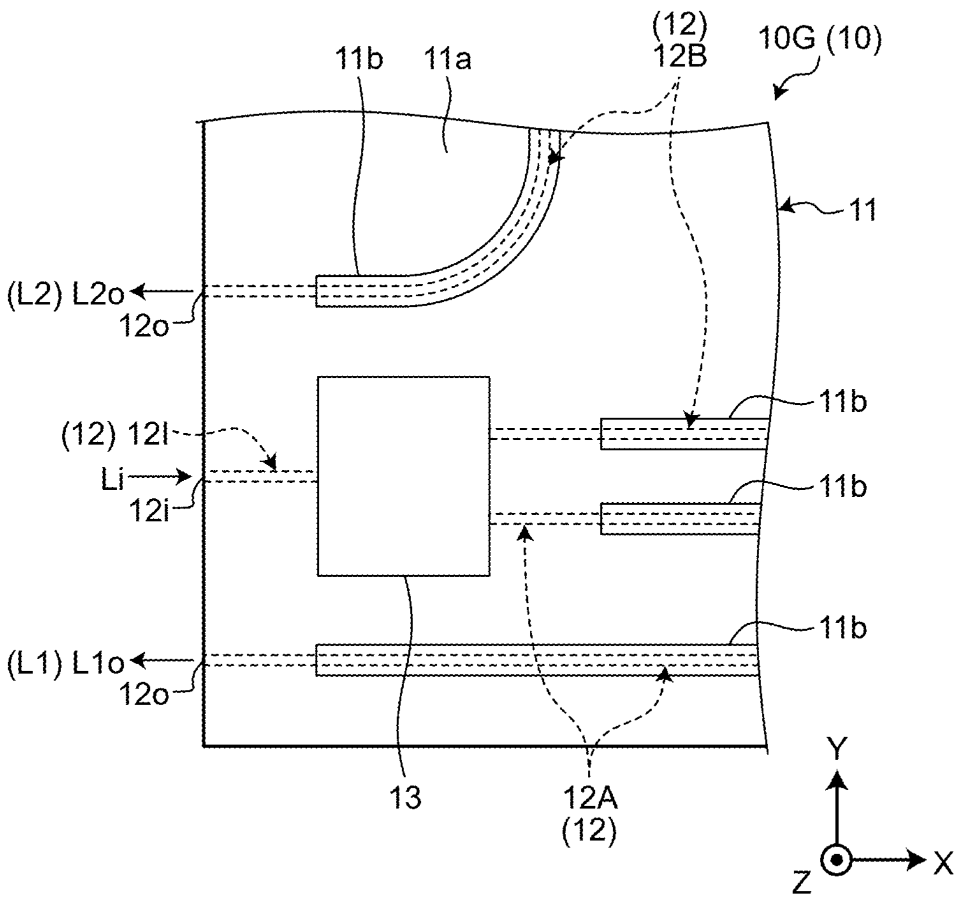
10E (10)
(11b)
11bb
11a
11
S
11b1
T1
12B
(12)
Wb
Z
Y
X

FIG.14



(12)
12B
11b
11b
10F (10)
11a
11
11b
11b
(L2) L2o
(L2) L2i
11b
11b
(L1) L1i
(L1) L1o
12A
(12)
11b
Y
X
Z

FIG.15



(12)
12B
10G (10)
11b
11a
11
(L2) L2o
12o
11b
(12) 12I
Li
12i
11b
11b
(L1) L1o
12o
13
12A
(12)
Y
Z
X

FIG.16

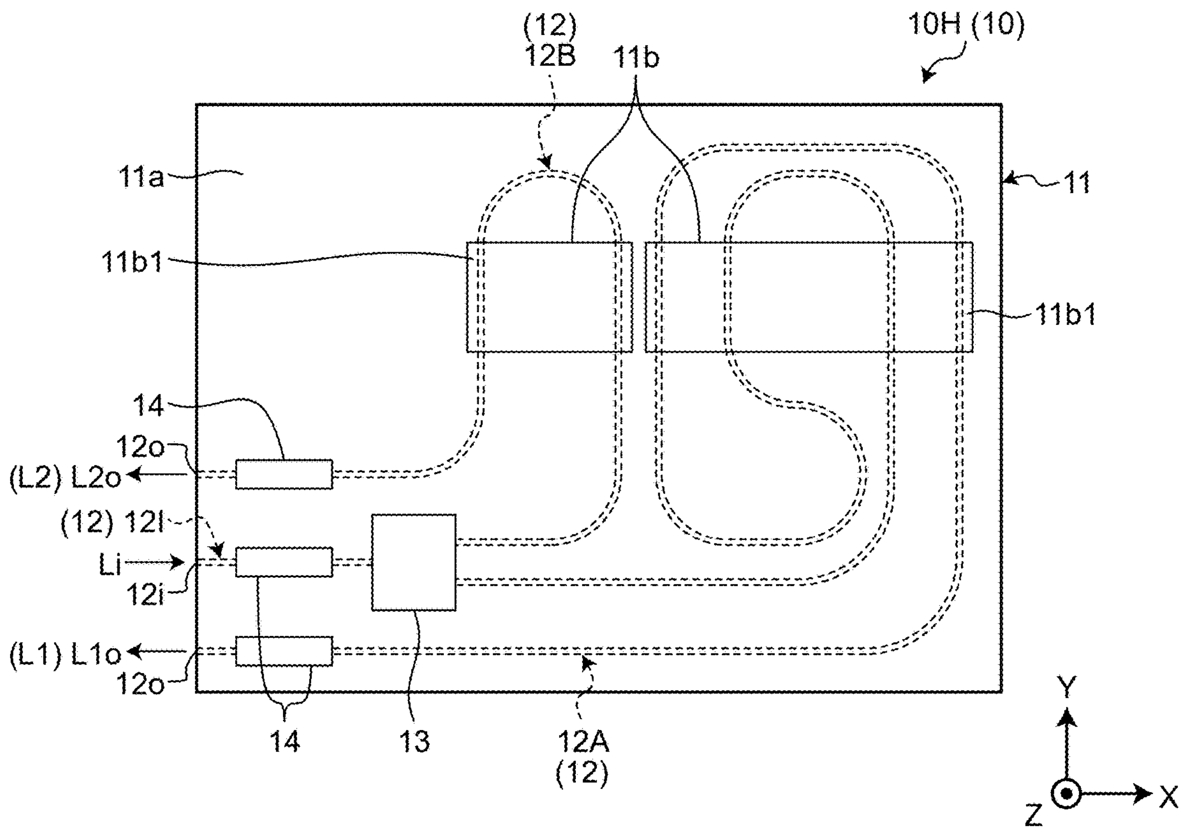
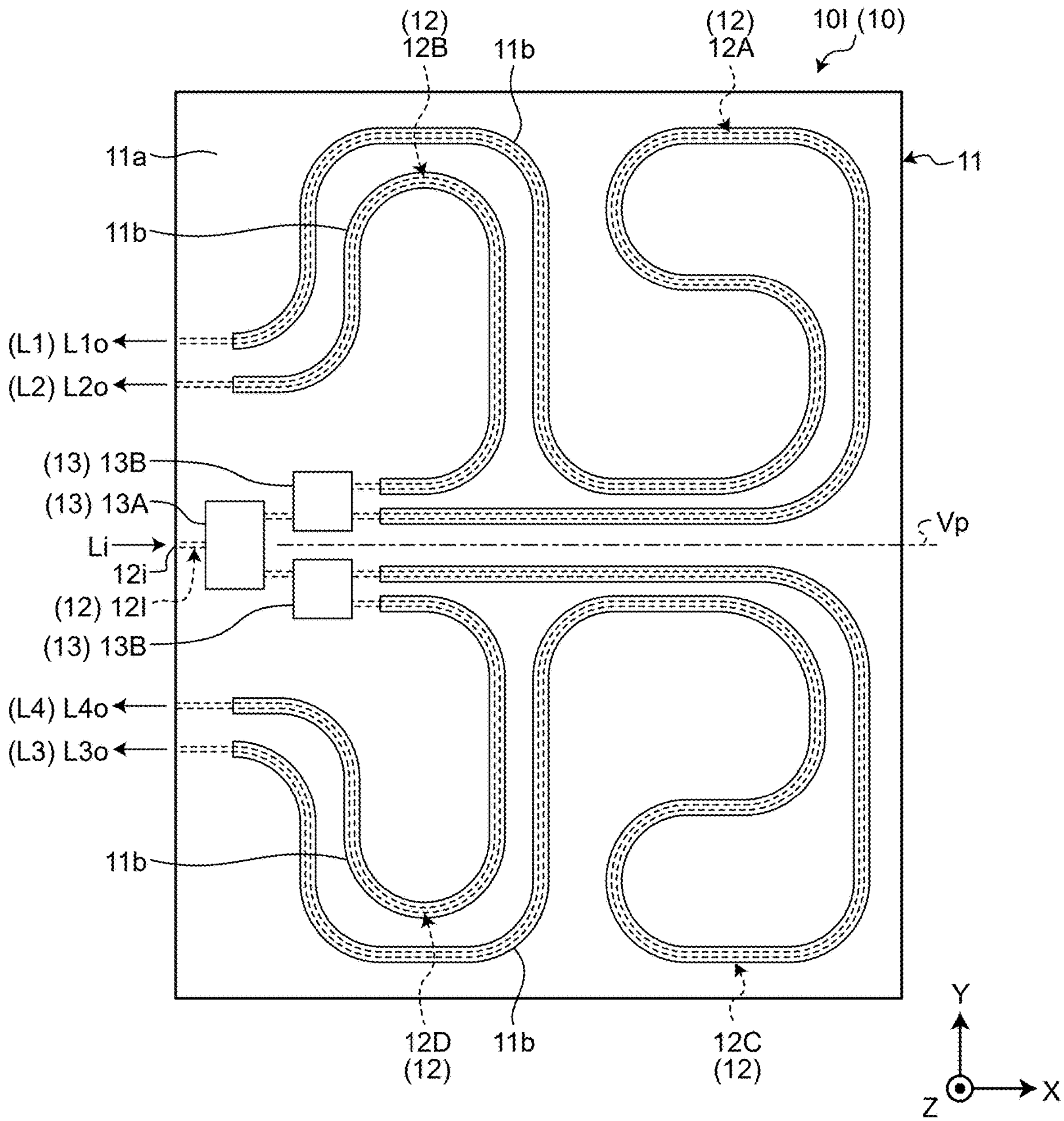

(12)
12B
11b
10H (10)
11a
11
11b1
11b1
14
12o
(L2) L2o
(12) 12I
Li
12i
(L1) L1o
12o
14
13
12A
(12)
Y
Z
X

FIG.17



(12)
12B
11b
(12)
12A
10I (10)
11
11a
11b
(L1) L1o
(L2) L2o
(13) 13B
(13) 13A
Li
12i
(12) 12I
(13) 13B
Vp
(L4) L4o
(L3) L3o
11b
12D
(12)
11b
12C
(12)
Y
X
Z

FIG.18

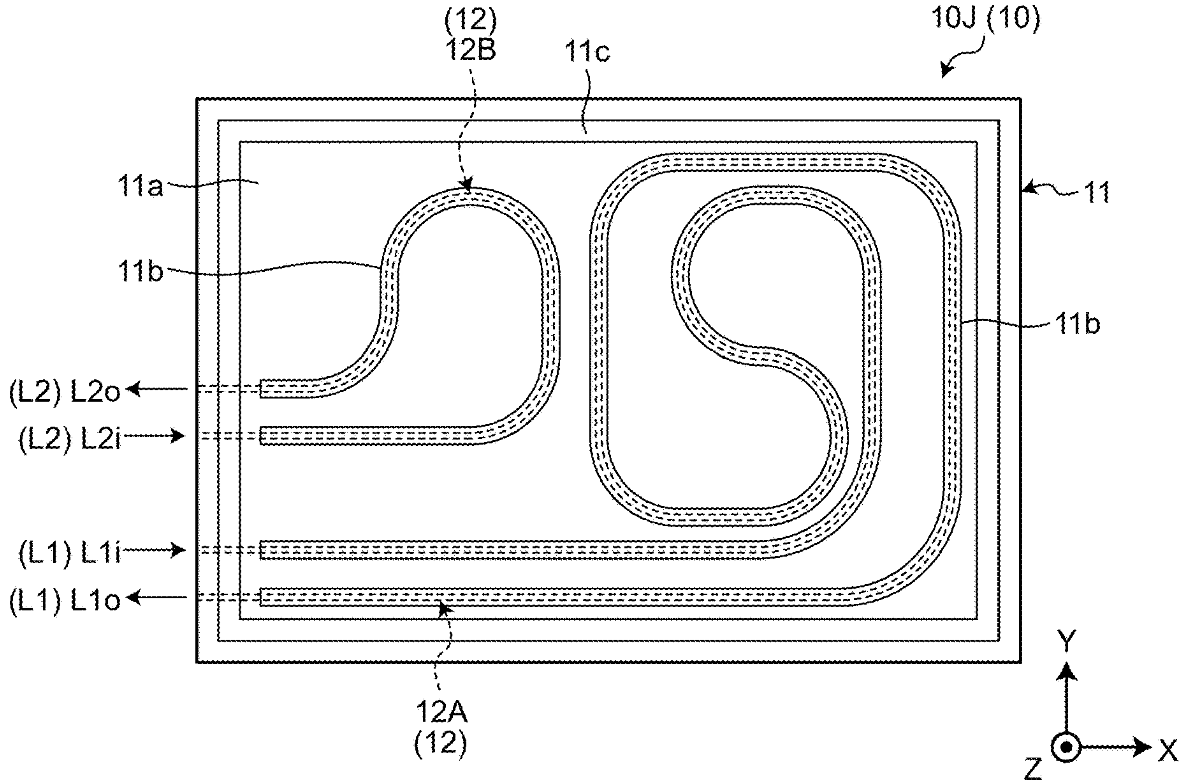

(12)
12B
11c
10J (10)
11a
11
11b
11b
(L2) L2o
(L2) L2i
(L1) L1i
(L1) L1o
12A
(12)
Y
X
Z

FIG.19

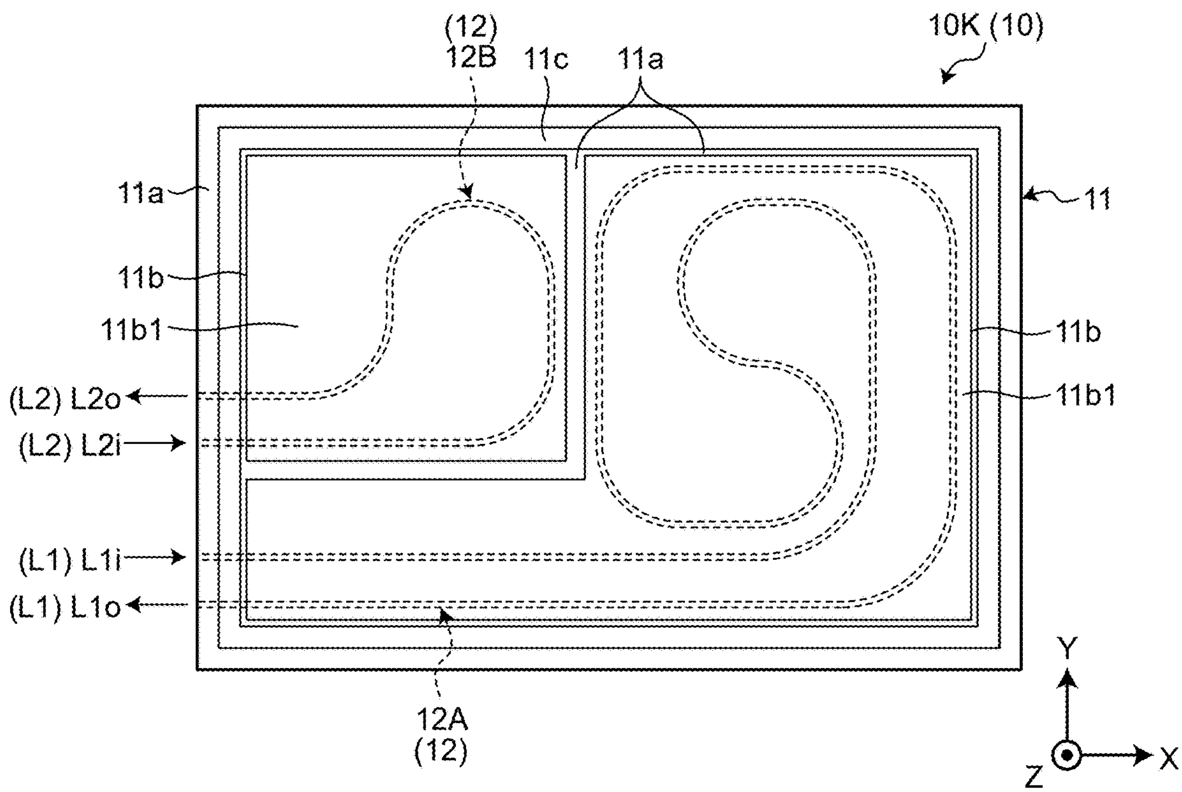

(12)
12B
11c
11a
10K (10)
11a
11
11b
11b1
11b
11b1
(L2) L2o
(L2) L2i
(L1) L1i
(L1) L1o
12A
(12)
Y
X
Z

FIG.20

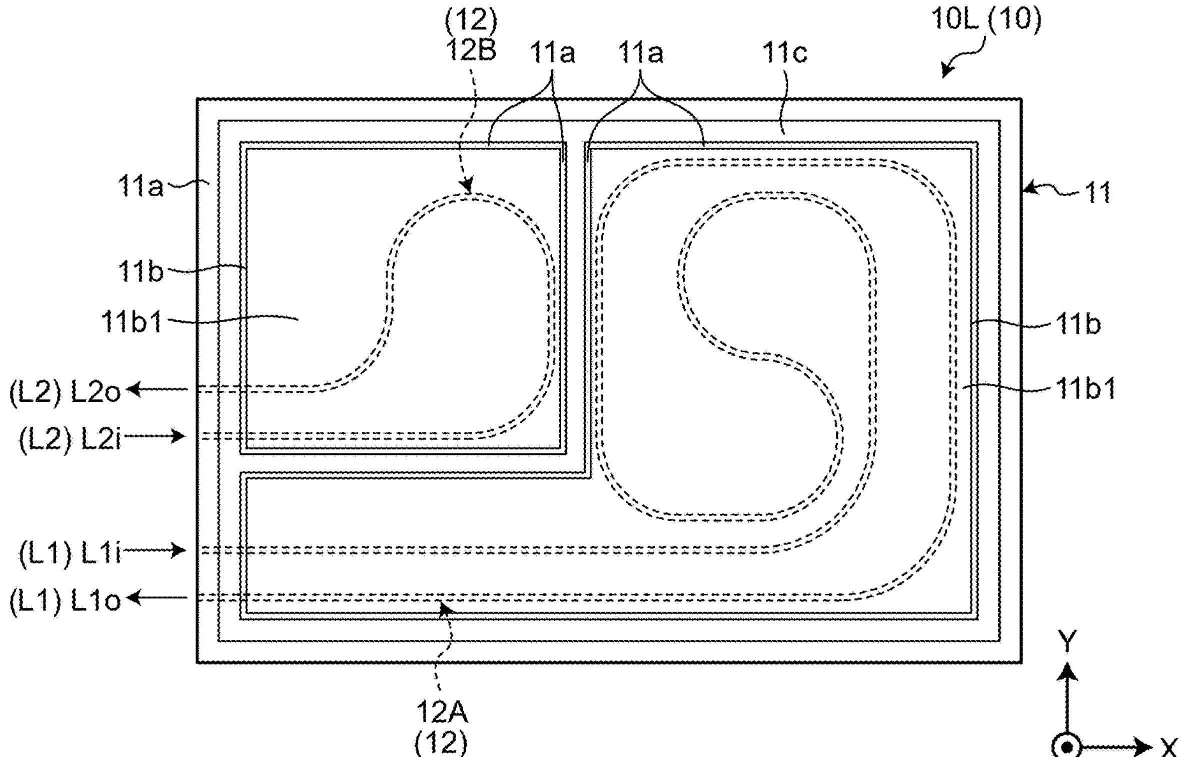

(12)
12B
11a
11a
11c
10L (10)
11a
11
11b
11b1
11b
11b1
(L2) L2o
(L2) L2i
(L1) L1i
(L1) L1o
12A
(12)
Y
Z
X

FIG.21

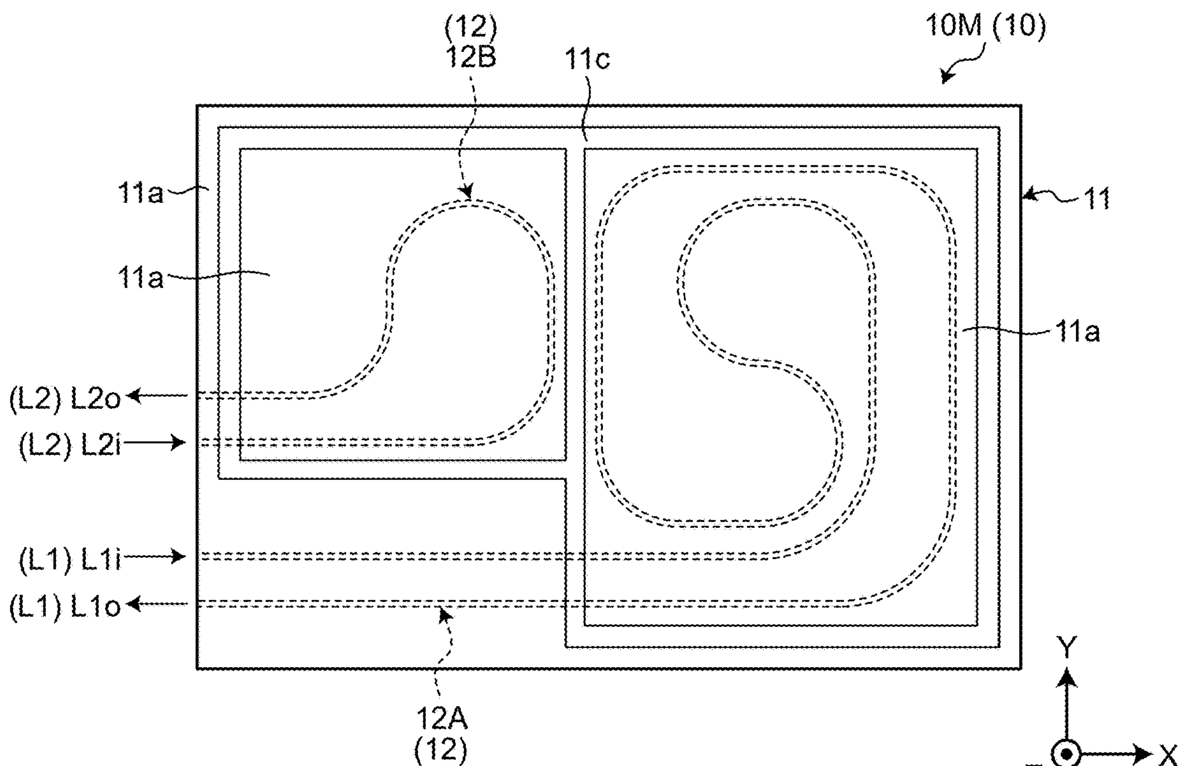

(12)
12B
11c
10M (10)
11a
11
11a
11a
(L2) L2o
(L2) L2i
(L1) L1i
(L1) L1o
12A
(12)
Y
Z
X

FIG.22

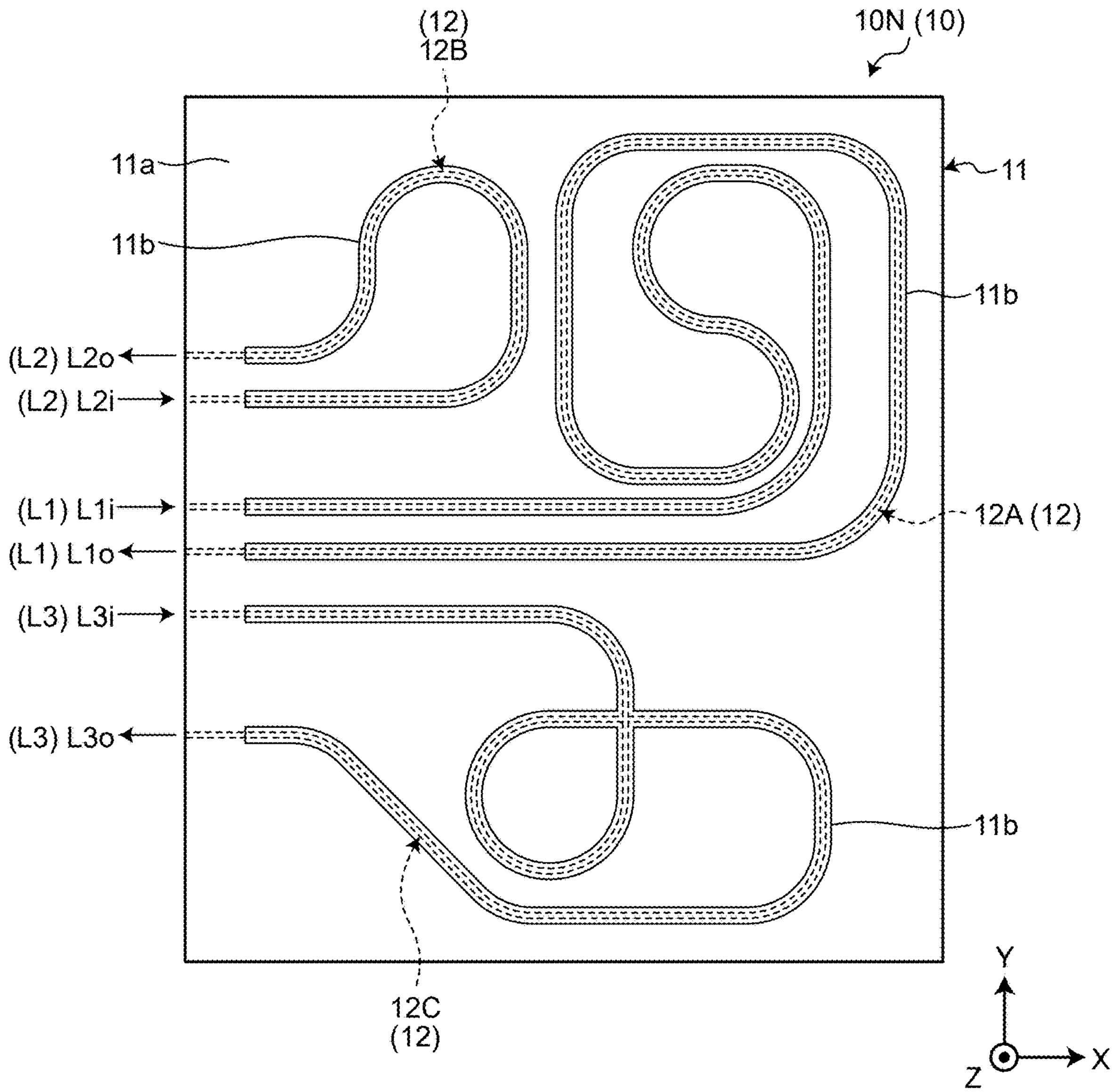

(12)
12B
10N (10)
11a
11
11b
11b
(L2) L2o
(L2) L2i
(L1) L1i
12A (12)
(L1) L1o
(L3) L3i
(L3) L3o
11b
12C
(12)
Y
Z
X

FIG.23

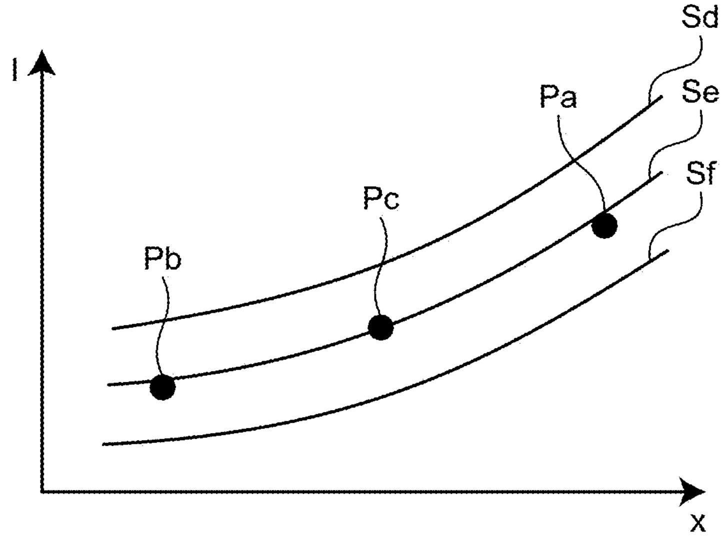
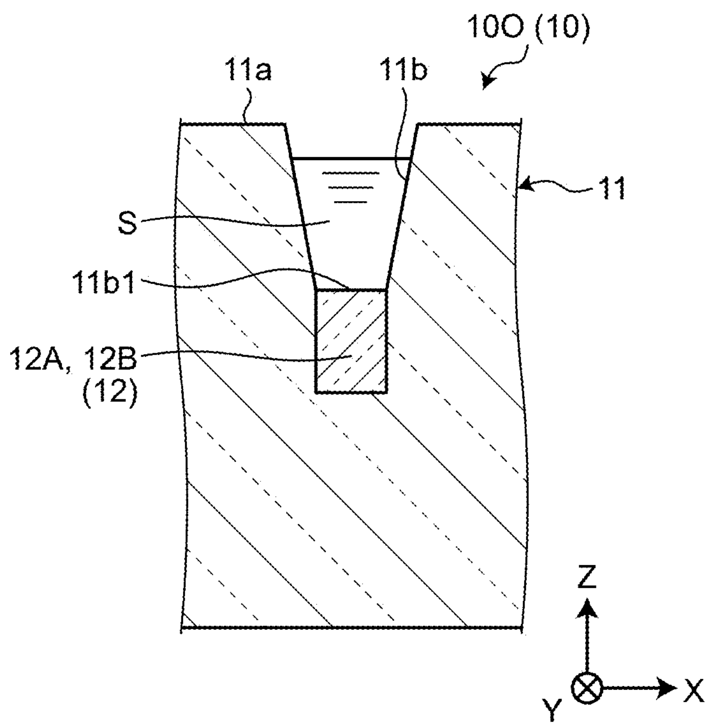


FIG.24



OPTICAL WAVEGUIDE SENSOR

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of International Application No. PCT/JP2024/037273, filed on Oct. 18, 2024 which claims the benefit of priority of the prior Japanese Patent Application No. 2023-181432, filed on Oct. 20, 2023, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present disclosure relates to an optical waveguide sensor.

2. Description of the Related Art

[0003] In the related art, a sensor is known that measures the components or the concentration of a material using the ATR method (ATR stands for Attenuated Total Reflection) (for example, Japanese Patent Application Laid-open No. 2005-61904). When there occurs total reflection of infrared light inside an ATR crystal prism, the specimen that is making contact with the prism can be examined by the sensor using the light that sparsely seeps to the outside of the prism (i.e., the evanescent light).

SUMMARY OF THE INVENTION

[0004] In such types of sensors, for example, if the examination accuracy can be further enhanced, it would prove beneficial.

[0005] In that regard, it is desirable to provide a new and improved optical waveguide sensor that, for example, enables achieving further enhancement in the examination accuracy.

[0006] In some embodiments, an optical waveguide sensor includes: a plurality of first waveguides, each first waveguide including a core at least partially enclosed by a cladding and being configured to guide an examination light; and a contact surface with which an examination subject is in contact and from which leaks seeping component of the examination light guided by the first waveguides. The examination light guided by the plurality of first waveguides has mutually different amount of leakage from the contact surface.

[0007] In some embodiments, an optical waveguide sensor includes: a plurality of first waveguides, each first waveguide including a core at least partially enclosed by a cladding and being configured to guide an examination light; and a contact surface with which an examination subject is in contact and from which leaks seeping component of the examination light guided by the first waveguides. The contact surface includes a first contact surface corresponding to each of the first waveguides.

[0008] In some embodiments, an optical waveguide sensor includes: a plurality of first waveguides, each first waveguide including a core at least partially enclosed by a cladding and being configured to guide an examination light; a contact surface with which an examination subject is in contact and from which leaks seeping component of the examination light guided by the first waveguides; and an identification structure that enables visual identification of the contact surface.

[0009] The above and other objects, features, advantages and technical and industrial significance of this disclosure will be better understood by reading the following detailed description of presently preferred embodiments of the disclosure, when considered in connection with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. **1** is an illustrative and schematic planar view diagram of an optical waveguide sensor according to a first embodiment;

[0011] FIG. **2** is a cross-sectional view of an II-II position in the optical waveguide sensor illustrated in FIG. **1**;

[0012] FIG. **3** is a cross-sectional view at an IIIa-IIIa position and an IIIb-IIIb position in the optical waveguide sensor illustrated in FIG. **1**;

[0013] FIG. **4** is a diagram illustrating an example of the regression analysis of a parameter and the loss intensity of the detection result obtained by the optical waveguide sensor according to the first embodiment;

[0014] FIG. **5** is an illustrative and schematic planar view diagram of an optical waveguide sensor according to a second embodiment;

[0015] FIG. **6** is a VI-VI cross-sectional view of the optical waveguide sensor illustrated in FIG. **5**;

[0016] FIG. **7** is a VII-VII cross-sectional view of the optical waveguide sensor illustrated in FIG. **5**;

[0017] FIG. **8** is an illustrative and schematic planar view diagram of an optical waveguide sensor according to a third embodiment;

[0018] FIG. **9** is cross-sectional view at an IXa-IXa position and an IXb-IXb position of the optical waveguide sensor illustrated in FIG. **8**;

[0019] FIG. **10** is an illustrative and schematic planar view diagram of an optical waveguide sensor according to a fourth embodiment;

[0020] FIG. **11** is an illustrative and schematic planar view diagram of an optical waveguide sensor according to a fifth embodiment;

[0021] FIG. **12** is an XII-XII cross-sectional view of the optical waveguide sensor illustrated in FIG. **11**;

[0022] FIG. **13** is an XIII-XIII cross-sectional view of the optical waveguide sensor illustrated in FIG. **11**;

[0023] FIG. **14** is an illustrative and schematic planar view diagram of an optical waveguide sensor according to a sixth embodiment;

[0024] FIG. **15** is an illustrative and schematic planar view diagram of some part of an optical waveguide sensor according to a seventh embodiment;

[0025] FIG. **16** is an illustrative and schematic planar view diagram of an optical waveguide sensor according to an eighth embodiment;

[0026] FIG. **17** is an illustrative and schematic planar view diagram of an optical waveguide sensor according to a ninth embodiment;

[0027] FIG. **18** is an illustrative and schematic planar view diagram of an optical waveguide sensor according to a 10-th embodiment;

[0028] FIG. **19** is an illustrative and schematic planar view diagram of an optical waveguide sensor according to an 11-th embodiment;

[0029] FIG. **20** is an illustrative and schematic planar view diagram of an optical waveguide sensor according to a 12-th embodiment;

[0030] FIG. **21** is an illustrative and schematic planar view diagram of an optical waveguide sensor according to a 13-th embodiment;

[0031] FIG. **22** is an illustrative and schematic planar view diagram of an optical waveguide sensor according to a 14-th embodiment;

[0032] FIG. **23** is a diagram illustrating an example of the regression analysis of a parameter and the loss intensity of the detection result obtained by the optical waveguide sensor according to the 14-th embodiment; and

[0033] FIG. **24** is an illustrative and schematic cross-sectional view, at an equivalent position to FIG. **3**, of an optical waveguide sensor according to a modification example of the first embodiment.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0034] Exemplary embodiments of the disclosure are described below. The configurations explained in the embodiments described below as well as the actions and the results (effects) attributed to the configurations are only exemplary. Thus, the disclosure can be implemented also using some different configuration than the configurations disclosed in the embodiments described below. Meanwhile, according to the disclosure, it becomes possible to achieve at least one of various effects (including secondary effects) that are attributed to the configurations.

[0035] The embodiments described below include identical constituent elements. Thus, based on the identical configuration according to each embodiment, it becomes possible to achieve identical actions and identical effects. In the following explanation, the identical constituent elements are referred to by the same reference numerals, and their explanation is not given in a repeated manner.

[0036] The drawings are schematic diagrams, and the dimensions illustrated in the drawings are sometimes different than the dimensions of the actual objects. In the drawings, the X direction is indicated by an arrow X, the Y direction is indicated by an arrow Y, and the Z direction is indicated by an arrow Z. The X direction, the Y direction, and the Z direction intersect with each other and are orthogonal to each other. In the present written description, the planar view indicates the line of sight viewed from the opposite direction of the Z direction, and is a planar view diagram represents a diagram in the planar view.

First Embodiment

[0037] FIG. **1** is a planar view diagram of an optical waveguide sensor **10**A (**10**) according to a first embodiment. For example, the optical waveguide sensor **10** has a relatively thinner and flattened cuboid shape in the Z direction. At the end portion of the optical waveguide sensor **10** in the Z direction, a surface **11***a* is present that surfaces the Z direction, that intersects with the Z direction, and that is substantially orthogonal to the Z direction. Herein, the Z direction represents an example of a first direction, and the surface **11***a* represents an example of a first surface.

[0038] The optical waveguide sensor **10** can be configured as, for example, a known planar lightwave circuit (PLC). In that case, the optical waveguide sensor **10** includes, in an integrated manner, a substrate that expands to intersect with the Z direction and a structure that is laminated on the substrate in the Z direction. The substrate is a glass substrate or a silicon substrate. The structure on the substrate includes cores **12** and a cladding **11** that encloses the cores **12**. The cladding **11** is made of, for example, a silica based glass material containing SiO_2; and the cores **12** are made of a silica based glass material having higher refractive index than the refractive index of the cladding **11**.

[0039] In the optical waveguide sensor **10**, the cores **12** transmit an examination light. Accompanying the transmission of the examination light; the seeping component of the examination light, that is, the evanescent light gets distributed in those portions of the cladding **11** which are in the vicinity of the cores **12**. Herein, the cores **12** represent an example of waveguides (first waveguides).

[0040] As illustrated in FIG. **1**, the optical waveguide sensor **10** includes a plurality of cores **12** (**12**A and **12**B), that is, includes a plurality of waveguides. Each core **12** extends in a curved manner and substantially along a virtual surface intersecting with the Z direction. An examination light L**1** that is input to an inlet end **12***i* of the core **12**A is transmitted along the core **12**A, and is output from an outlet end **12***o* of the core **12**A. Similarly, an examination light L**2** that is input to the inlet end **12***i* of the core **12**B is transmitted along the core **12**B, and is output from the outlet end **12***o* of the core **12**B. In the first embodiment, L**1***i* represents the incident light of the examination light L**1**, and L**1***o* represents the outgoing light of the examination light L**1**. Similarly, L**2***i* represents the incident light of the examination light L**2**, and L**2***o* represents the outgoing light of the examination light L**2**.

[0041] As illustrated in FIG. **1**, in the first embodiment, the core **12**A and the core **12**B have mutually different lengths, and the length of the core **12**A is greater than the length of the core **12**B.

[0042] FIG. **2** is a cross-sectional view of an II-II position in the optical waveguide sensor **10**A illustrated in FIG. **1**. At the II-II position, in the surrounding portion of the core **12**A as well as the surrounding portion of the core **12**B, the optical waveguide sensor **10**A has the cross-sectional shape illustrated in FIG. **2**, that is, has the identical cross-sectional shape. Moreover, in the first embodiment, between the inlet end **12***i* and the outlet end **12***o*, each core **12** has the same cross-sectional shape as the shape illustrated in FIG. **2**.

[0043] FIG. **3** is a cross-sectional view at an IIIa-IIIa position and an IIIb-IIIb position in the optical waveguide sensor **10**A illustrated in FIG. **1**. In the surrounding portion of the core **12**A as well as the surrounding portion of the core **12**B, the optical waveguide sensor **10**A has the cross-sectional shape illustrated in FIG. **3**, that is, has the identical cross-sectional shape. Meanwhile, in the first embodiment, although the cores **12** (**12** and **12**B) have a quadrangular cross-sectional shape as an example, that is not the only possible case. Thus, the cross-sectional shape can be any other shape other than the quadrangular shape.

[0044] As illustrated in FIG. **3**, in the optical waveguide sensor **10**, recesses **11***b* are formed that are recessed in the opposite direction to the Z direction from the surface **11***a* toward the cores **12**. In the optical waveguide sensor **10**, as illustrated in FIG. **1**, the recesses **11***b* are formed at a total of two positions corresponding to the cores **12**A and **12**B. The two recesses **11***b* are disposed to overlap with the cores **12**A and **12**B in the Z direction, and extend along the corresponding cores **12**A and **12**B in the entire section excluding the vicinity of the inlet end **12***i* and the outlet end **120**. Moreover, the recesses **11***b* extend, in the respective entire sections, in the cross-sectional shape illustrated in FIG. **3**, that is, extend with a substantially constant width and a substantially constant depth. A bottom surface **11***b***1** of each recess **11***b* is a flat surface that is oriented toward the Z direction, that intersects with the Z direction, and that expands in the substantially orthogonal direction to the Z direction. Each bottom surface **11***b***1** substantially overlaps with the corresponding core **12** in the Z direction. Moreover, each bottom surface **11***b***1** has the substantially same width as the width of the corresponding core **12**. However, that is not the only possible case. Thus, the width of each bottom surface **11***b***1** can be greater or smaller than the width of the corresponding core **12**. Moreover, the width of the bottom surfaces **11***b***1** is substantially constant along the longitudinal direction of the corresponding recesses **11***b*. However, that is not the only possible case. Alternatively, the bottom surfaces **11***b***1** can have locally different widths in different portions.

[0045] In such a configuration, an examination subject S is placed inside each recess **11***b*. The examination subject S is, for example, a liquid, a gas, a fluid, a viscoelastic object, or a solid object having flexibility. As explained above, in the portions of the cladding **11** that surround the cores **12**, the evanescent light of the examination lights L**1** and L**2** gets distributed. In between each core **12** and the corresponding bottom surface **11***b***1** of the recess **11***b*, a thickness T**1** of the cladding **11** is relatively smaller such as equal to or smaller than 5 [μm]. When the thickness T**1** is smaller than the mode field radius of the examination lights L**1** and L**2** passing through the cores **12**, the mode field of the examination lights L**1** and L**2** spreads out more toward the inside of the recesses **11***b* than toward the bottom surfaces **11***b***1** thereof. Hence, the evanescent light of the examination lights L**1** and L**2** leaks from the bottom surfaces **11***b***1** of the recesses **11***b*, and gets absorbed in the examination subject S. In that case, the intensity of the intensity lights L**1***o* and L**2***o* decreases with respect to the intensity of the incident lights L**1***i* and L**2***i* by the amount equivalent to the amount of absorption of the examination lights L**1** and L**2** by the examination subject S. The amount of decrease (the loss intensity) of the outgoing lights L**1***o* and L**2***o* with respect to the incident lights L**1***i* and L**2***i*, that is, the amount of absorption differs according to the examination subject S. Hence, measuring the loss intensity enables identification of the examination subject S. Each bottom surface **11***b***1** represents an example of a contact surface with which the examination subject S is in contact, and each recess **11***b* represents an example of a receiving recess and a first recess. Moreover, the thickness T**1** represents an example of the distance from the cores **12**A and **12**B to the bottom surfaces **11***b***1**. In the cross-sectional surface illustrated in FIG. **2**, since the cladding **11** around the cores **12**A and **12**B is thick, there is no leakage of the evanescent light. Meanwhile, the thickness T**1** is equal to or greater than "0" (T**1**≥0).

[0046] As illustrated in FIG. **1**, in the two recesses **11***b* corresponding to the cores **12**A and **12**B, the bottom surfaces **11***b***1** have mutually different lengths. As explained earlier, in the two recesses **11***b*, the bottom surfaces **11***b***1** have the same width that is substantially constant along the longitudinal direction. Thus, in each of the two recesses **11***b*, the area of the bottom surface **11***b***1** becomes equal to (constant width)×(length); and that area differs according to the length, that is, differs in proportion to the length. In the first embodiment, the length of the recess **11***b* corresponding to the core **12**A is greater than the length of the recess **11***b* corresponding to the core **12**B. Hence, the area of the bottom surface **11***b***1** of the recess **11***b* corresponding to the core **12**A becomes greater than the area of the bottom surface **11***b***1** of the recess **11***b* corresponding to the core **12**B.

[0047] The amount of leakage of the evanescent light from each bottom surface **11***b***1** differs according to the area of that bottom surface **11***b***1** and increases in proportion to the area. Thus, in the optical waveguide sensor **10**A (**10**) according to the first embodiment, two waveguides are included that have different amounts of leakage of the evanescent light from the respective bottom surfaces **11***b***1**.

[0048] FIG. **4** is a diagram illustrating an example of the regression analysis of a parameter x and a loss intensity I of the detection result obtained by the optical waveguide sensor **10**. In the graph illustrated in FIG. **4**, the parameter x on the horizontal axis represents the parameter for varying the amount of leakage of the examination light from the waveguide. For example, the parameter x represents the area of the length of the bottom surface **11***b***1**. The loss intensity I on the vertical axis represents the loss intensity of the examination lights L**1** and L**2** corresponding to the cores **12**A and **12**B (the waveguides). In FIG. **4** is illustrated the experiment result corresponding to the same examination lights L**1** and L**2**. In the graph, a point Pa represents the coordinates of the measurement result of the parameter x and the loss intensity I corresponding to the core **12**A, and a point Pb represents the coordinates of the measurement result of the parameter x and the loss intensity I corresponding to the core **12**B. The bottom surface **11***b***1** of the recess **11***b* corresponding to the core **12**A has a greater area (or a greater length) than the area (or the length) of the bottom surface **11***b***1** of the recess **11***b* corresponding to the core **12**B. Hence, the value of the parameter x at the point Pa is greater than the value of the parameter x at the point Pb. Moreover, greater the value of the parameter x, the greater becomes the loss intensity I. Hence, the loss intensity I at the point Pa is greater than the loss intensity I at the point Pb. In that case, for example, if a function obtained by performing regression analysis of the points Pa and Pb is compared with a function of the parameter x and the loss intensity I of each of known materials Sa to Sc, it becomes possible to identify whether the examination subject S is same as any one of the materials Sa to Sc or to figure out the material from among the materials Sa to Sc to which the properties of the examination subject S are close. Meanwhile, in FIG. **4** is illustrated the example in which the parameter x and the loss intensity I are proportional to each other. However, the parameter x need not be proportional to the loss intensity I.

[0049] As explained above, the optical waveguide sensor **10**A (**10**) includes two waveguides (first waveguides) that have different amounts of leakage of the evanescent light from the corresponding bottom surfaces **11***b***1** (contact surfaces). With such a configuration, the two parameters x can be measured and the two loss intensities I corresponding to the parameters x can be measured; regression analysis can be performed, for example, using two combinations of the parameters x and the loss intensities I; and the examination subject S can be identified and the properties of the examination subject S can be figured out. That is, according to the first embodiment, as a result of measuring the loss intensity in each of the two waveguides that have different amounts of leakage of the evanescent light, the examination of the examination subject S can be performed with more accuracy as compared to the case of measuring the loss intensity in only a single waveguide.

[0050] Moreover, in the first embodiment, in the optical waveguide sensor **10**, the recesses **11***b* (the receiving recesses) are provided in which the examination subject S is placed, and the bottom surfaces **11***b***1** of the recesses **11***b* serve as the contact surfaces onto which the evanescent light leaks. With such a configuration, as compared to a configuration in which no recess is provided, the examination subject S can be held in the measurable state with more ease and more stability.

[0051] Moreover, in the first embodiment, in the optical waveguide sensor **10**, a plurality of recesses **11***b* is provided. With such a configuration, for example, mutually different examination subjects S can be placed in the recesses **11***b*, and the examination can be carried out for the different examination subjects S.

Second Embodiment

[0052] FIG. **5** is a planar view diagram of an optical waveguide sensor **10**B (**10**) according to a second embodiment. As illustrated in FIG. **5**, in the second embodiment, two cores **12**A and **12**B (**12**) are in the mirror image relationship having plane symmetry with respect to a virtual plane Vp that intersects with the Y direction. Moreover, the recesses **11***b* that are provided corresponding to the cores **12**A and **12**B have the same length but have mutually different shapes.

[0053] FIG. **6** is a VI-VI cross-sectional view of the optical waveguide sensor **10**B illustrated in FIG. **5**, and FIG. **7** is a VII-VII cross-sectional view of the optical waveguide sensor **10**B illustrated in FIG. **5**. In the second embodiment too, in an identical manner to the first embodiment, the recesses **11***b* corresponding to the cores **12**A and **12**B extend along the cores **12**A and **12**B, respectively, with the cross-sectional shapes illustrated in FIGS. **6** and **7**, respectively. When FIG. **6** is compared with FIG. **7**, it is clear that the depth of the recess **11***b* (see FIG. **6**), which corresponds to the core **12**A, from the surface **11***a* is greater than the depth of the recess **11***b* (see FIG. **7**), which corresponds to the core **12**B, from the surface **11***a*. Accordingly, a thickness T**21** (T**2**, see FIG. **6**) between the core **12**A and the bottom surface **11***b***1** of the recess **11***b* is smaller than a thickness T**22** (T**2**, see FIG. **7**) between the core **12**B and the bottom surface **11***b***1** of the recess **11***b*. Hence, the amount of leakage of the evanescent light from the bottom surface **11***b***1** in the optical waveguide including the core **12**A is greater than the amount of leakage of the evanescent light from the bottom surface **11***b***1** in the optical waveguide including the core **12**B. That is, in the second embodiment too, the optical waveguide sensor **10**B (**10**) includes two waveguides having different amounts of leakage of the evanescent light from the corresponding bottom surfaces **11***b***1**. Thus, in the second embodiment, by treating the thicknesses T**2** (T**21** and T**22**) as the parameters x, regression analysis can be performed in an identical manner to the first embodiment as illustrated in FIG. **4**. The thicknesses T**2** (T**21** and T**22**) represent examples of the distances from the cores **12**A and **12**B to the corresponding bottom surfaces **11***b***1**.

[0054] In this way, in the second embodiment too, the optical waveguide sensor **10**B (**10**) includes two waveguides (first waveguides) that have different amounts of leakage of the evanescent light from the corresponding bottom surfaces **11***b***1** (contact surfaces). Thus, in the second embodiment too, as a result of measuring the loss intensity in each of the two waveguides, the examination of the examination subject S can be performed with more accuracy as compared to the case of measuring the loss intensity in only a single waveguide.

Third Embodiment

[0055] FIG. **8** is a planar view diagram of an optical waveguide sensor **10**C (**10**) according to a third embodiment. When FIG. **8** is compared with FIG. **1**, it is clear that the optical waveguide sensor **10**C according to the third embodiment includes the cores **12**A and **12**B (**12**) having the same shape as in the optical waveguide sensor **10**A according to the first embodiment.

[0056] However, as illustrated in FIG. **8**, in the third embodiment, unlike the recesses **11***b* extending in an elongated manner along the cores **12**A and **12**B according to the first and second embodiments described above, the recesses **11***b* are formed to be wider than the cores **12**A and **12**B and have a rectangular shape in the planar view.

[0057] FIG. **9** is cross-sectional view at an IXa-IXa position and an IXb-IXb position of the optical waveguide sensor **10**C illustrated in FIG. **8**. At the IXa-IXa position and the IXb-IXb position, in the optical waveguide sensor **10**C, although the recesses **11***b* and the bottom surfaces **11***b***1** have a different size, the cross-sectional shape is identical as illustrated in FIG. **9**. As illustrated in FIG. **9**, the recesses **11***b* and the bottom surfaces **11***b***1** are wider than the cores **12**A and **12**B in the Y direction. That is, the bottom surface **11***b***1** includes first areas A**1** that overlap with the cores **12**A and **12**B in the Z direction, and includes second areas A**2** that do not overlap with the cores **12**A and **12**B in the Z direction but are positioned adjacent to the first areas A**1**. In the third embodiment too, in an identical manner to the first embodiment, the thickness T**1** between the cores **12** and the corresponding bottom surfaces **11***b***1** is relatively smaller such as equal to or smaller than 5 [μm]. Thus, in the third embodiment, of the bottom surfaces **11***b***1**, the evanescent light leaks mainly from the first areas A**1**. Herein, the first areas A**1** represent examples of a contact surface.

[0058] The optical waveguide sensor **10**C is configured in such a way that, in the portions in which the cores **12**A and **12**B overlap with the recesses **11***b*, the cross-sectional shape is identical to FIG. **9**. Hence, in the third embodiment, of the cores **12**A and **12**B, the evanescent light leaks from the sections overlapping with the recesses **11***b*. As illustrated in FIG. **8**, the length of the section in which the core **12**A overlaps with the corresponding bottom surface **11***b***1** (hereinafter, referred to as a first section) is different than the length of the section in which the core **12**B overlaps with the corresponding bottom surface **11***b***1** (hereinafter, referred to as a second section). In the third embodiment, as an example, the length of the first section is greater than the length of the second section. Thus, in the third embodiment, the amount of leakage of the evanescent light from the bottom surface **11***b***1** in the waveguide including the core **12**A is greater than the amount of leakage of the evanescent light from the bottom surface **11***b***1** in the waveguide including the core **12**B. That is, in the third embodiment too, the optical waveguide sensor **10**C (**10**) includes two waveguides having different amounts of leakage of the evanescent light from the corresponding bottom surfaces **11***b***1**. Thus, in the third embodiment, if the lengths over which the cores **12**A and **12**B overlap with the corresponding bottom surfaces **11***b***1** are treated as the parameters x, regression analysis can be performed in an identical manner to the first embodiment as illustrated in FIG. **4**. Alternatively, the projected areas of the cores **12**A and **12**B onto the corresponding bottom surfaces **11***b***1** in the Z direction can be treated as the parameters x. That is, the values of multiplication of the length and the width of the cores **12**A and **12**B can be treated as the parameters x.

[0059] In this way, in the third embodiment too, the optical waveguide sensor **10**C (**10**) includes two waveguides (first waveguides) that have different amounts of leakage of the evanescent light from the corresponding bottom surfaces **11***b***1** (contact surfaces). Thus, in the third embodiment too, as a result of measuring the loss intensity in each of the two waveguides having different amounts of leakage of the evanescent light, the examination of the examination subject S can be performed with more accuracy as compared to the case of measuring the loss intensity in only a single waveguide.

[0060] Moreover, in the third embodiment, the bottom surfaces **11***b***1** are wider on account of including the first areas A**1** and the second areas A**2**. When the recesses **11***b* are narrow in width, the examination subject S that is relatively harder cannot be examined. As a result, the examination subject S cannot easily enter into the recess **11***b* and cannot make contact with the bottom surface **11***b***1**. In other words, when the recesses **11***b* are narrow in width, the examination subject S that can be examined gets confined to an object having fluidity or flexibility so that the object can enter into the recess **11***b* and can make contact with the bottom surface **11***b***1**. In that regard, in the third embodiment, since the recesses **11***b* are wide, even the examination subject S having less fluidity or less flexibility can make contact with the bottom surface **11***b***1**, and by extension can make contact with that area of the bottom surface **11***b***1** from which the evanescent light leaks (in the third embodiment, mainly the first areas A**1**). Hence, such an examination subject S can also be examined. Herein, the first areas A**1** represent examples of a contact surface. Meanwhile, in the third embodiment, since the shape of the recesses **11***b* is simplified, sometimes it becomes possible to reduce the time and the cost required in manufacturing the optical waveguide sensor **10**C.

Fourth Embodiment

[0061] FIG. **10** is a planar view diagram of an optical waveguide sensor **10**D (**10**) according to a fourth embodiment. In the optical waveguide sensor **10**D according to the fourth embodiment, a single recess **11***b* is formed. The recess **11***b* is formed when the cladding **11** between the two recesses **11***b* formed in the optical waveguide sensor **10**C according to the third embodiment is removed and the two recesses **11***b* are integrated. Other than that, the optical waveguide sensor **10**D has an identical configuration to the optical waveguide sensor **10**C according to the third embodiment.

[0062] According to the fourth embodiment too, it becomes possible to achieve the same effects as the effects achieved according to the third embodiment. Moreover, according to the fourth embodiment, as a result of achieving further simplification in the shape of the recess **11***b*, sometimes it becomes possible to further reduce the time and the cost required in manufacturing the optical waveguide sensor **10**D.

Fifth Embodiment

[0063] FIG. **11** is a planar view diagram of an optical waveguide sensor **10**E (**10**) according to a fifth embodiment. The optical waveguide sensor **10**E according to the fifth embodiment has a partially identical configuration to the optical waveguide sensor **10**B (see FIG. **5**). More particularly, the core **12**A according to the fifth embodiment has the same shape as the core **12**A according to the second embodiment. Moreover, in the planar view, the core **12**B according to the fifth embodiment extends in the same shape as the core **12**B according to the second embodiment. However, in the fifth embodiment, the core **12**B includes a plurality of sections having different widths. More particularly, the core **12**B includes a first section **12***a* having a narrow width, a second section **12***b* that is wider than the width of the first section **12***a*, and a third section **12***c* that undergoes gradual variation in the width along the longitudinal direction in between the first section **12***a* and the second section **12***b*. The recess **11***b* includes a narrow-width portion **11***ba* corresponding to the first section **12***a*, a wide portion **11***bb* corresponding to the second section **12***b*, and an intermediate portion **11***bc* corresponding to the third section **12***c*.

[0064] FIG. **12** is an XII-XII cross-sectional view of the optical waveguide sensor **10**E illustrated in FIG. **11**, and FIG. **13** is an XIII-XIII cross-sectional view of the optical waveguide sensor **10**E. As illustrated in FIGS. **12** and **13**, the recess **11***b* is positioned to overlap with the core **12**B in the Z direction, and extends along the core **12**B. The depth of the recess **11***b* is constant regardless of the position of the core **12**B, and is same as the depth of the recess **11***b* corresponding to the core **12**A. Hence, the thickness T**1** between the core **12**B and the corresponding bottom surface **11***b***1** is constant across all sections of the recess **11***b*, and is same as the thickness T**1** between the core **12**A and the corresponding bottom surface **11***b***1**. For example, the thickness T**1** is equal to or smaller than 5 [μm]. Hence, the evanescent light leaks from the substantially entire bottom surface **11***b***1** of the recess **11***b* corresponding to the core **12**B. Meanwhile, the bottom surface **11***b***1** of the recess **11***b* corresponding to the core **12**B substantially overlaps with the core **12**B in the Z direction. That is, the width of the bottom surface **11***b***1** of the narrow-width portion **11***ba* and the width of the bottom surface **11***b***1** of the wide portion **11***bb* are substantially same as a width Wa and a width Wb of the corresponding core **12**B, respectively, and changes according to the variation occurring in the width Wa and the width Wb of the core **12**B, respectively.

[0065] In that case, the area of the bottom surface **11***b***1** corresponding to the core **12**B becomes greater than the area of the bottom surface **11***b***1** corresponding to the core **12**A. Thus, the amount of leakage of the evanescent light from the bottom surface **11***b***1** in the waveguide including the core **12**B becomes greater than the amount of leakage of the evanescent light from the bottom surface **11***b***1** in the waveguide including the core **12**A. That is, in the fifth embodiment too, the optical waveguide sensor **10**E (**10**) includes two waveguides having different amounts of leakage of the evanescent light from the corresponding bottom surfaces **11***b***1**. Thus, in the fifth embodiment, by treating the areas of the bottom surfaces **11***b***1** as the parameters x, regression analysis can be performed in an identical manner to the first embodiment as illustrated in FIG. **4**.

[0066] In this way, in the fifth embodiment too, the optical waveguide sensor **10**E (**10**) includes two waveguides (first waveguides) that have different amounts of leakage of the evanescent light from the corresponding bottom surfaces **11***b***1** (contact surfaces). Thus, in the fifth embodiment too, as a result of measuring the loss intensity in each of the two waveguides, the examination of the examination subject S can be performed with more accuracy as compared to the case of measuring the loss intensity in only a single waveguide.

Sixth Embodiment

[0067] FIG. **14** is a planar view diagram of an optical waveguide sensor **10**F (**10**) according to a sixth embodiment. When FIG. **14** is compared with FIG. **1**, it is clear that the cores **12**A and **12**B (**12**) of the optical waveguide sensor **10**F according to the sixth embodiment have the same shape as the cores **12**A and **12**B of the optical waveguide sensor **10**A according to the first embodiment. However, in the sixth embodiment, in each of the cores **12** and **12**B, the recesses **11***b* are provided at a plurality of positions in a dispersed manner. Each recess **11***b* has an identical cross-sectional shape as the shape according to the first embodiment (see FIG. **3**), extends along the corresponding core from among the cores **12**A and **12**B, and has a constant width. Moreover, the cores **12**A and **12**B include straight sections extending in a linear manner in the planar view and include curved sections that are curved in the planar view. From among the straight sections and the curved sections, the recesses **11***b* are formed corresponding to only the straight sections. The total length of the recesses **11***b* corresponding to the core **12**A is greater than the total length of the recesses **11***b* corresponding to the core **12**B. The bottom surface **11***b***1** (see FIG. **3**) of each recess **11***b* represents an example of a contact surface and represents an example of a first contact surface. Meanwhile, the recesses **11***b* need not be formed in all straight sections.

[0068] In the sixth embodiment too, the evanescent light leaks from the bottom surface **11***b***1** of each recess **11***b*. Thus, in the sixth embodiment too, the optical waveguide sensor **10**F (**10**) includes two waveguides (first waveguides) that have different amounts of leakage of the evanescent light from the corresponding bottom surfaces **11***b***1** (contact surfaces). Thus, in the sixth embodiment too, as a result of measuring the loss intensity in each of the two waveguides, the examination of the examination subject S can be performed with more accuracy as compared to the case of measuring the loss intensity in only a single waveguide.

[0069] Moreover, in the curved sections of the cores **12**A and **12**B, sometimes there is an increase in the leakage of the evanescent light due to the curved shape. Accompanying such increase, there is a risk of a decline in the examination accuracy. In that regard, according to the sixth embodiment, it becomes possible to eliminate the impact of the leakage of the evanescent light occurring due to the curved shape of the curved sections. As a result, the examination of the examination subject S can be performed with greater accuracy. In other words, according to the sixth embodiment, in each of the cores **12**A and **12**B (**12**), a plurality of bottom surfaces **11***b***1** (contact surfaces) can be placed at favorable positions in a dispersed manner. As a result, the examination accuracy can be enhanced while securing the amount of leakage of the evanescent light required for the examination. Seventh embodiment

[0070] FIG. **15** is a planar view diagram of some part of an optical waveguide sensor **10**G (**10**) according to a seventh embodiment. The configuration according to the seventh embodiment can be substituted for the configuration in the vicinity of the inlet end **12***i* and the outlet end **120** of the cores **12**A and **12**B (**12**) of the optical waveguide sensor **10**A according to the first embodiment.

[0071] The optical waveguide sensor **10**G according to the seventh embodiment includes a branching portion **13**. The branching portion **13** branches an examination light L**1**, which is input from the inlet end **12***i* of a core **121** (**12**), to the two cores **12**A and **12**B. The core **12**I represents an example of a core of a second waveguide.

[0072] The branching portion **13** can be configured as, for example, a splitter. In that case, examples of the branching portion **13** include a power splitter, a wavelength division multiplexing (WDM) splitter, a polarization beam splitter, and a variable beam splitter.

[0073] When the branching portion **13** is a power splitter, it splits the examination light Li to the core **12**A and **12**B at the already-set intensity ratio (for example, 1:1).

[0074] When the branching portion **13** is a WDM splitter, it splits the examination light Li to the cores **12**A and **12**B for each different wavelength band. In that case, for each different wavelength band, the absorption property of the examination subject S can be examined according to the loss intensity I in each of the cores **12**A and **12**B.

[0075] When the branching portion **13** is a polarization beam splitter, it splits the examination light Li into the transverse electric (TE) polarization component and the transverse magnetic (TM) polarization component. In that case, for each polarization component, the absorption property of the examination subject S can be examined according to the loss intensity I in each of the cores **12**A and **12**B.

[0076] When the branching portion **13** is a variable beam splitter, it can variably set the intensity ratio to be used in splitting the examination light Li to the cores **12**A and **12**B.

[0077] Moreover, the branching portion **13** can be configured as, for example, an optical switch. In that case, the branching portion **13** can selectively input the examination Li to either the core **12**A or the core **12**B in a time-sharing manner. The optical switch represents an example of a switch.

[0078] In either case, in the optical waveguide sensor **10**G (**10**), as a result of including the branching portion **13**, the number of light sources can be reduced as compared to the case in which the branching portion **13** is not included. Moreover, a variety of examinations can be performed using a relatively simpler configuration.

Eighth Embodiment

[0079] FIG. **16** is a planar view diagram of an optical waveguide sensor **10**H (**10**) according to an eighth embodiment. As illustrated in FIG. **16**, the optical waveguide sensor **10**H according to the eighth embodiment includes the branching portion **13** (see FIG. **13**) identical to the seventh embodiment. Hence, according to the eighth embodiment, it becomes possible to achieve identical effects to the effects achieved according to the seventh embodiment.

[0080] In the optical waveguide sensor **10**H according to the eighth embodiment, relatively wider recesses **11***b* (see

FIGS. **8** and **9**) identical to the third embodiment are formed corresponding to each of the cores **12**A and **12**B. Hence, according to the eighth embodiment, it is possible to achieve identical effects to the effects achieved according to the third embodiment.

[0081] Moreover, according to the eighth embodiment, in an identical manner to the sixth embodiment, the recesses **11***b* are formed corresponding to the straight sections of the cores **12**A and **12**B (see FIG. **14**). Hence, according to the eighth embodiment, it is possible to achieve identical effects to the effects achieved according to the sixth embodiment.

[0082] Moreover, in the eighth embodiment, a mode filter **14** that removes the high-order mode components from the examination light is disposed in each of the cores **12**A, **12**B, and **121**. According to the eighth embodiment, as a result of being able to remove the high-order mode components not required in the examination, the examination accuracy can be further enhanced. The mode filter **14** can be used in the configurations according to the other embodiments too. Meanwhile, the installation positions of the mode filters **14** are not limited to the positions illustrated in FIG. **16**.

Ninth Embodiment

[0083] FIG. **17** is a planar view diagram of an optical waveguide sensor **101** (**10**) according to a ninth embodiment. As illustrated in FIG. **17**, the optical waveguide sensor **10**I according to the ninth embodiment includes branching portions **13**A and **13**B (**13**) and cores **12**A, **12**B, **12**C, **12**D, and **12**I (**12**). The examination light Li that is input to the inlet end **12***i* of the core **121** gets branched via the branching portions **13**A and **13**B (**13**).

[0084] The cores **12**A and **12**C are in the mirror image relationship having plane symmetry with respect to the virtual plane Vp that intersects with the Y direction. Similarly, the cores **12**B and **12**D are in the mirror image relationship having plane symmetry with respect to the virtual plane Vp. Moreover, the recess **11***b* corresponding to the core **12**A and the recess **11***b* corresponding to the core **12**C are in the mirror image relationship having plane symmetry with respect to the virtual plane Vp. Similarly, the recess **11***b* corresponding to the core **12**B and the recess **11***b* corresponding to the core **12**D are in the mirror image relationship having plane symmetry with respect to the virtual plane Vp. Between each of the cores **12**A to **12**D and the bottom surface **11***b***1** of the corresponding recess **11***b*, the thickness T**1** (see FIG. **3**) is same and remains constant without variation in the longitudinal direction.

[0085] The branching portion **13**A is, for example, a polarization beam splitter; and the branching portion **13**B is, for example, a splitter that splits the light at the intensity ratio of 1:1. In that case, for example, the properties of the examination subject S with respect to the TE polarization component can be examined from the loss intensity I in the waveguides respectively including the cores **12**A and **12**B; and the properties of the examination subject S with respect to the TM polarization component can be examined from the loss intensity I in the waveguides respectively including the cores **12**C and **12**D. In this way, as a result of including the branching portions **13**, a plurality of properties of the examination subject S can be examined using the examination light Li output from only a single light source. Eventually, sometimes it becomes possible to enhance the examination accuracy regarding the examination subject.

10-th Embodiment to 12-th Embodiment

[0086] FIG. **18** is a planar view diagram of an optical waveguide sensor **10**J (**10**) according to a 10-th embodiment; FIG. **19** is a planar view diagram of an optical waveguide sensor **10K** (**10**) according to an 11-th embodiment; and FIG. **20** is a planar view diagram of an optical waveguide sensor **10**L (**10**) according to a 12-th embodiment.

[0087] As illustrated in FIGS. **18** to **20**, the optical waveguide sensors **10**J, **10**K, and **10**L according to the 10-th to 12-th embodiments include the cores **12**A and **12**B identical to the first embodiment.

[0088] As illustrated in FIG. **18**, in the optical waveguide sensor **10**J according to the 10-th embodiment, the recesses **11***b* are formed in an identical manner to the first embodiment. Hence, according to the 10-th embodiment, it becomes possible to achieve identical effects to the effects achieved according to the first embodiment.

[0089] Moreover, as illustrated in FIGS. **19** and **20**, in the optical waveguide sensors **10**K and **10**L according to the 11-th and 12-th embodiments, respectively; relatively wider recesses **11***b* (see FIG. **8**) are formed corresponding to each of the cores **12**A and **12**B in an identical manner to the third embodiment. Hence, according to the 11-th and 12-th embodiments, it is possible to achieve identical effects to the effects achieved according to the third embodiment.

[0090] Furthermore, as illustrated in FIGS. **18** to **20**, in the optical waveguide sensors **10**J, **10**K, and **10**L according to the 10-th to 12-th embodiments, respectively; a groove **11***c* is formed to enclose the recesses **11***b* in the planar view. The groove **11***c* is recessed from the surface **11***a* in the opposite direction to the Z direction and, for example, extends with a constant depth. However, in the portion passing across at least the cores **12**, the depth of the groove **11***c* is set in such a way that the groove **11***c* does not reach the cores **12**.

[0091] If the groove **11***c* is not formed in the optical waveguide sensors **10**J, **10**K, and **10**L and if the examination subject S is, for example, a liquid; when the examination subject S is poured in excess quantity in the recesses **11***b*, the examination subject S spills out from the surface **11***a* of the optical waveguide sensors **10**J, **10**K, and **10**L. In that regard, in the 10-th to 12-th embodiments, since the quantity of the examination subject S exceeding the volume of the recesses **11***b* gets collected in the groove **11***c*, the examination subject S can be held down from spilling around the optical waveguide sensors **10**J, **10**K, and **10**L. The groove **11***c* represents an example of a surrounding recess. As illustrated in FIGS. **18** and **19**, the groove **11***c* either can collectively enclose the recesses **11***b* and their bottom surfaces **11***b***1** (see FIG. **3**), or can individually enclose each recess **11***b* and its bottom surface **11***b***1**.

13-th Embodiment

[0092] FIG. **21** is a planar view diagram of an optical waveguide sensor **10**M (**10**) according to a 13-th embodiment. In the optical waveguide sensor **10**M according to the 13-th embodiment, the cores **12**A and **12**B are disposed in an identical manner to the first embodiment. However, in the optical waveguide sensor **10**M, the recesses **11***b* are not formed corresponding to the cores **12**A and **12**B. Moreover, between the cores **12**A and **12**B and the surface **11***a*, the cladding **11** is configured to have a constant thickness that is relatively thinner such as equal to or smaller than, for example, 5 [μm] in entirety.

[0093] Moreover, in the optical waveguide sensor **10**M according to the 13-th embodiment, the groove **11**c is formed in an identical manner to the 10-th to 12-th embodiments.

[0094] As explained above, in the 13-th embodiment, between the cores **12**A and **12**B and the surface **11**a, the thickness of the cladding **11** is relatively thinner. For that reason, of the surface **11**a, the evanescent light leaks from the regions overlapping with the cores **12**A and **12**B in the Z direction, that is, leaks from the elongated regions extending along the cores **12**A and **12**B, respectively. Thus, when the examination subject S is placed on such a region, the examination can be performed in an identical manner to the first embodiment. In that case, of the surface **11**a, the regions overlapping with the cores **12**A and **12**B in the Z direction represent examples of a contact surface.

[0095] In the configuration explained above, in order to perform the examination with accuracy, it is necessary to accurately manage the area of the contact surface corresponding to each of the cores **12**A and **12**B, that is, the area of each contact surface with which the examination subject S is in contact and on which the evanescent light leaks. In that regard, in the 13-th embodiment, the groove **11**c is used in managing the size of the contact surface. More particularly, of the surface **11**a, the examination subject S can be placed over the entire region of the inside of the groove **11**c for establishing the contact. In that case, of the surfaces **11**a, the linear regions overlapping with those sections of the cores **12**A and **12**B which are enclosed by the groove **11**c in the planar view serve as the contact surfaces. Meanwhile, for example, when the examination subject S is a liquid, depending on its surface tension, the entire liquid can be placed in the region enclosed by the groove **11**c of the surface **11**a. In that case, the liquid that spills out from the region enclosed by the groove **11**c gets collected in the groove **11**c. The groove **11**c represents an example of an identification structure that enables visual identification of the position serving as the contact surface for the user, that is, the position at which the examination subject S is to be placed.

[0096] According to the 13-th embodiment, the optical waveguide sensor **10**M (**10**) can be implemented using a simpler configuration. Moreover, because of the groove **11**c, the position of placing the examination subject S becomes physically or visually determinable with ease, thereby enabling achieving enhancement in the examination accuracy. Meanwhile, also in a configuration in which the groove **11**c is replaced with a wall protruding from the surface **11**a in the Z direction, identical effects can be achieved.

14-th Embodiment

[0097] FIG. **22** is a planar view diagram of an optical waveguide sensor **10**N (**10**) according to a 14-th embodiment. In the optical waveguide sensor **10**N according to the 14-th embodiment, in addition to including the cores **12**A and **12**B and the corresponding recesses **11**b in an identical manner to the optical waveguide sensor **10**A according to the first embodiment, the additional core **12**C and the recess **11**b corresponding to that core **12**C is also included. The core **12**C and the recess **11**b have the same cross-sectional shape as the first embodiment (see FIG. **3**) and, with the cross-sectional shape illustrated in FIG. **3**, extend to intersect with the Z direction. An incident light L**3**i of an examination light L**3** is input to the core **12**C; and an outgoing light L**30** of the examination light L**3** is output from the core **12**C.

[0098] FIG. **23** is a diagram illustrating an example of the regression analysis of the parameter x and the loss intensity I of the detection result obtained by the optical waveguide sensor **10**N. In FIG. **23** is illustrated the experiment result corresponding to the same examination lights L**1**, L**2**, and L**3**. In the graph, a point Pa represents the coordinates of the measurement result of the parameter x and the loss intensity I corresponding to the core **12**A; a point Pb represents the coordinates of the measurement result of the parameter x and the loss intensity I corresponding to the core **12**B; and a point Pc represents the coordinates of the measurement result of the parameter x and the loss intensity I corresponding to the core **12**C. In an identical manner to the first embodiment (see FIG. **4**), in the 14-th embodiment too, if a function obtained by performing regression analysis of the points Pa, Pb, and Pc is compared with a function of the parameter x and the loss intensity I of each of known materials Sd to Sf, it becomes possible to identify whether the examination subject S is same as any one of the materials Sd to Sf or to figure out the material from among the materials Sd to Sf to which the properties of the examination subject S are close. Meanwhile, in FIG. **23** is illustrated the example in which the parameter x and the loss intensity I are not proportional to each other. Alternatively, the parameter x can be proportional to the loss intensity I. Moreover, the function of the parameter x and the loss intensity I can be of various types.

Modification Example in Which Cladding Is Absent Between Core and Examination Subject

[0099] FIG. **24** is a cross-sectional view, at an equivalent position to FIG. **3**, of an optical waveguide sensor **100** according to a modification example of the first embodiment. Except for the configuration illustrated in FIG. **24**, the optical waveguide sensor **100** has an identical configuration to the first embodiment. In the surrounding portion of the core **12**A as well as the surrounding portion of the core **12**B, the optical waveguide sensor **100** has the cross-sectional shape illustrated in FIG. **24**, that is, has the identical cross-sectional shape. As illustrated in FIG. **24**, the cladding **11** can be absent between the cores **12** and the bottom surfaces **11**b**1** of the corresponding recesses **11**b. Thus, this configuration is obtained when the thickness T**1** in the configuration illustrated in FIG. **3** is set to “0”. With this configuration, the evanescent light of the examination lights L**1** and L**2** can be more reliably leaked onto the examination subject S.

[0100] While certain embodiments and modification examples have been described, these embodiments and modification examples have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions. Moreover, regarding the constituent elements, the specifications about the configurations and the shapes (structure, type, direction, shape, size, length, width, thickness, height, number, arrangement, position, material, etc.) can be suitably modified.

[0101] For example, the number of cores (waveguides) and the number of recesses can be equal to or greater than four. Moreover, an optical waveguide sensor can include a plurality of waveguides in which the amount of leakage is same. In that case, one of those waveguides can be used as the reference waveguide for the purpose of referring to the loss intensity of an examination subject having known properties.

[0102] Furthermore, the structure enabling identification of the contact surface can be in the form of characters or a design.

[0103] According to the disclosure, it becomes possible to provide a new and improved optical waveguide sensor that, for example, enables achieving enhancement in the measurement accuracy.

[0104] Although the disclosure has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

What is claimed is:

1. An optical waveguide sensor comprising:
- a plurality of first waveguides, each first waveguide including a core at least partially enclosed by a cladding and being configured to guide an examination light; and
- a contact surface with which an examination subject is in contact and from which leaks seeping component of the examination light guided by the first waveguides, wherein
- the examination light guided by the plurality of first waveguides has mutually different amount of leakage from the contact surface.

2. The optical waveguide sensor according to claim **1**, further comprising:
- a first surface that is positioned at an end portion in a first direction; and
- a receiving recess that is recessed from the first surface in an opposite direction to the first direction, the receiving recess being configured to receive the examination subject, wherein
- the plurality of first waveguides extend to intersect with the first direction, and
- the receiving recess has a bottom surface including the contact surface.

3. The optical waveguide sensor according to claim **2**, wherein
- the receiving recess includes a plurality of first recesses, and
- the bottom surface of each of the first recesses includes the contact surface present between a different first waveguide and the examination subject.

4. The optical waveguide sensor according to claim **2**, wherein the bottom surface includes
- a first region that overlaps with the core in the first direction, and
- a second region that does not overlap with the core in the first direction and that is positioned adjacent to the first region.

5. The optical waveguide sensor according to claim **1**, wherein the contact surface includes a plurality of first contact surfaces.

6. The optical waveguide sensor according to claim **1**, comprising an identification structure that enables visual identification of the contact surface.

7. The optical waveguide sensor according to claim **1**, further comprising:
- a first surface that is positioned at an end portion in a first direction; and
- a surrounding recess that is around the contact surface and that is recessed from the first surface in an opposite direction to the first direction.

8. The optical waveguide sensor according to claim **1**, wherein the first waveguides have mutually different lengths.

9. The optical waveguide sensor according to claim **1**, wherein the first waveguides have mutually different widths.

10. The optical waveguide sensor according to claim **1**, wherein the first waveguides have mutually different distances between the core and the contact surface.

11. The optical waveguide sensor according to claim **1**, further comprising:
- a second waveguide configured to guide the examination light; and
- a splitter configured to branch the examination light from the second waveguide to the first waveguides.

12. The optical waveguide sensor according to claim **1**, further comprising:
- a second waveguide configured to guide the examination light; and
- a switch configured to guide the examination light from the second waveguide to the first waveguides in a selective manner.

13. The optical waveguide sensor according to claim **1**, further comprising a mode filter configured to remove a high-order mode component from the examination light.

14. The optical waveguide sensor according to claim **1**, wherein the first waveguides extend in a curved manner.

15. The optical waveguide sensor according to claim **1**, wherein
- the first waveguides include
  - a straight section that extends in a linear manner and intersects with the first direction, and
  - a curved section that extends in a curved manner and intersects with the first direction, and
- the contact surface is provided corresponding to only the straight section in the straight section and the curved section.

16. The optical waveguide sensor according to claim **1**, wherein the optical waveguide sensor is made of a material containing SiO_2.

17. An optical waveguide sensor comprising:
- a plurality of first waveguides, each first waveguide including a core at least partially enclosed by a cladding and being configured to guide an examination light; and
- a contact surface with which an examination subject is in contact and from which leaks seeping component of the examination light guided by the first waveguides, wherein
- the contact surface includes a first contact surface corresponding to each of the first waveguides.

18. An optical waveguide sensor comprising:

a plurality of first waveguides, each first waveguide including a core at least partially enclosed by a cladding and being configured to guide an examination light;

a contact surface with which an examination subject is in contact and from which leaks seeping component of the examination light guided by the first waveguides; and

an identification structure that enables visual identification of the contact surface.

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